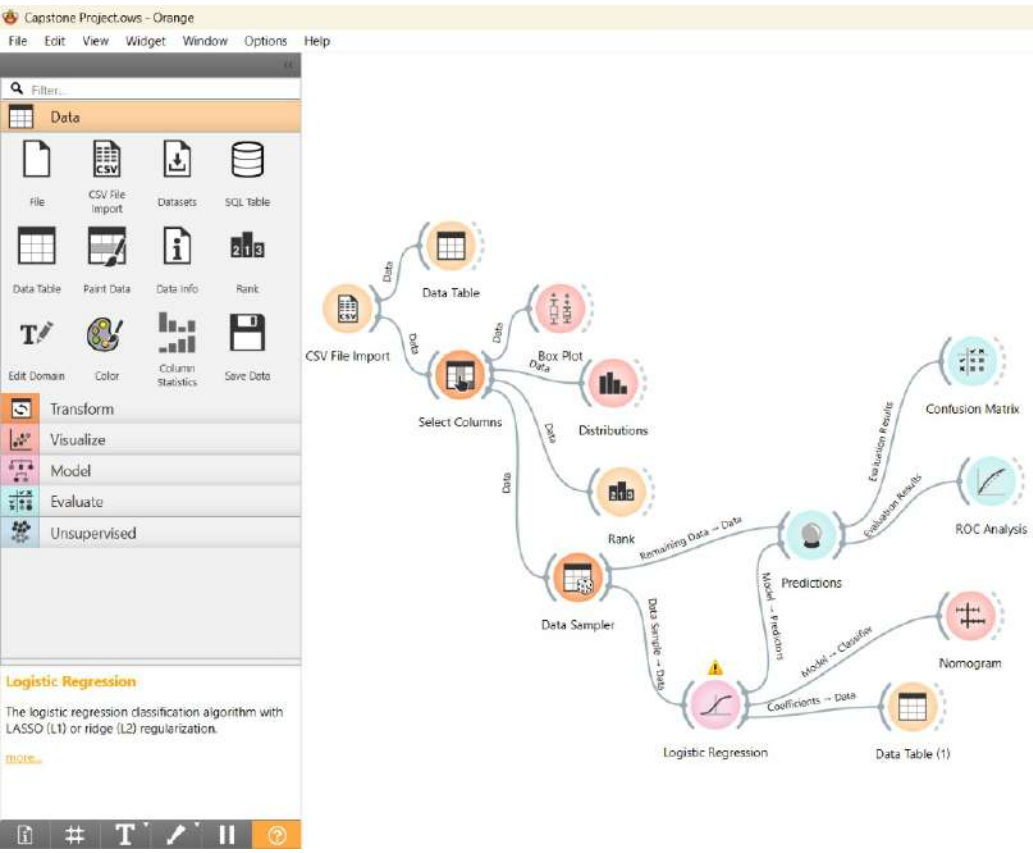
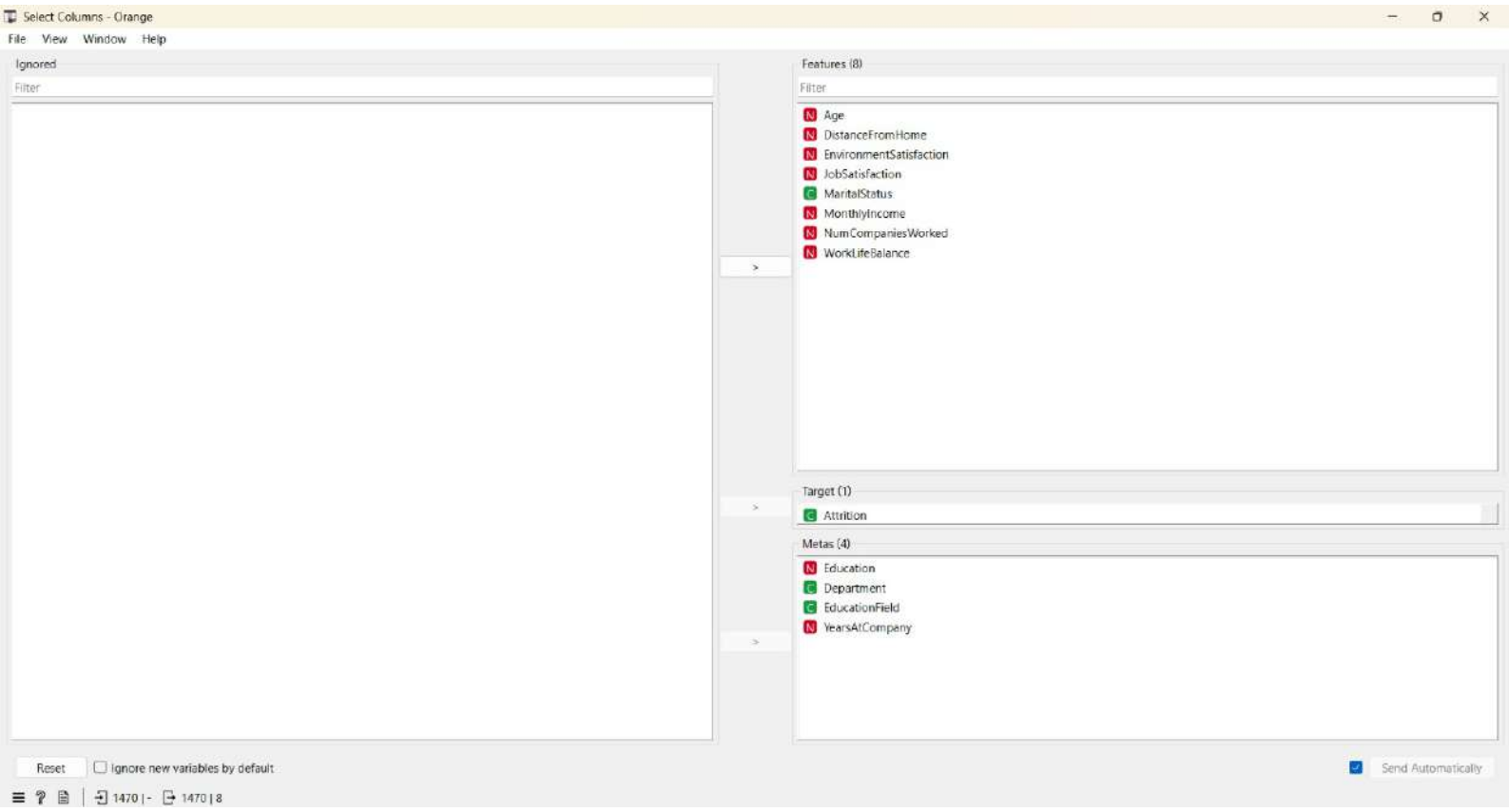
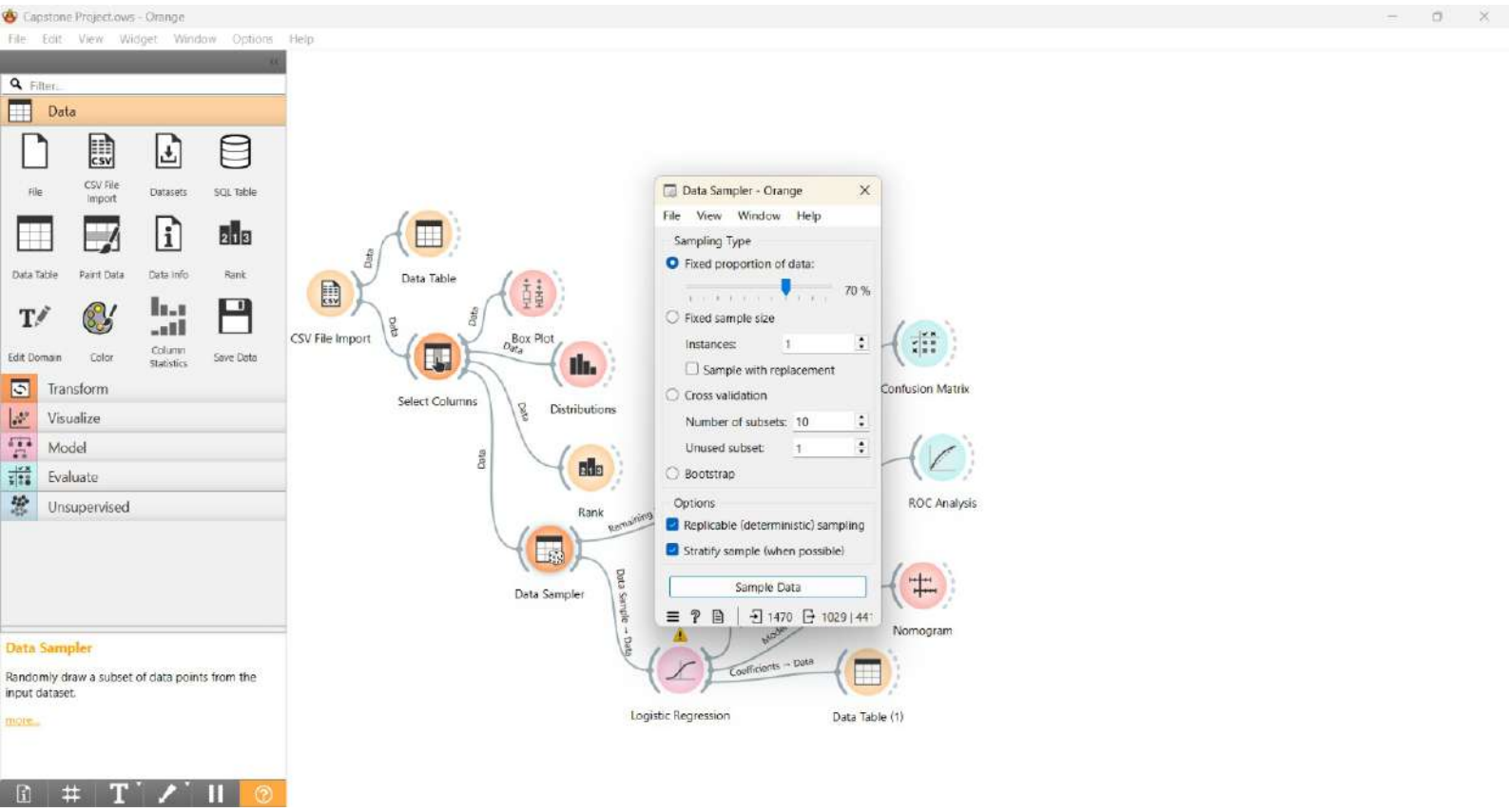
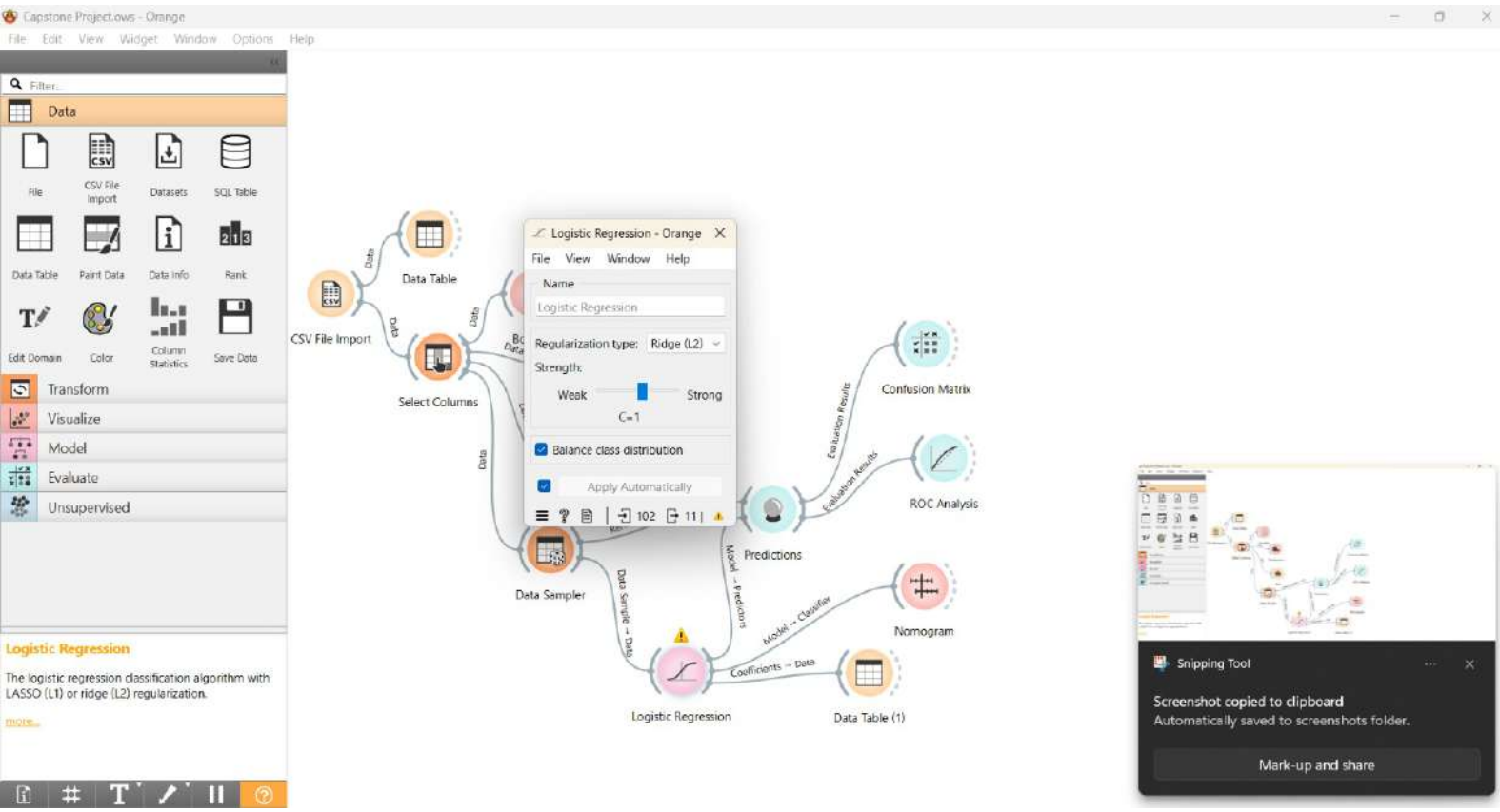
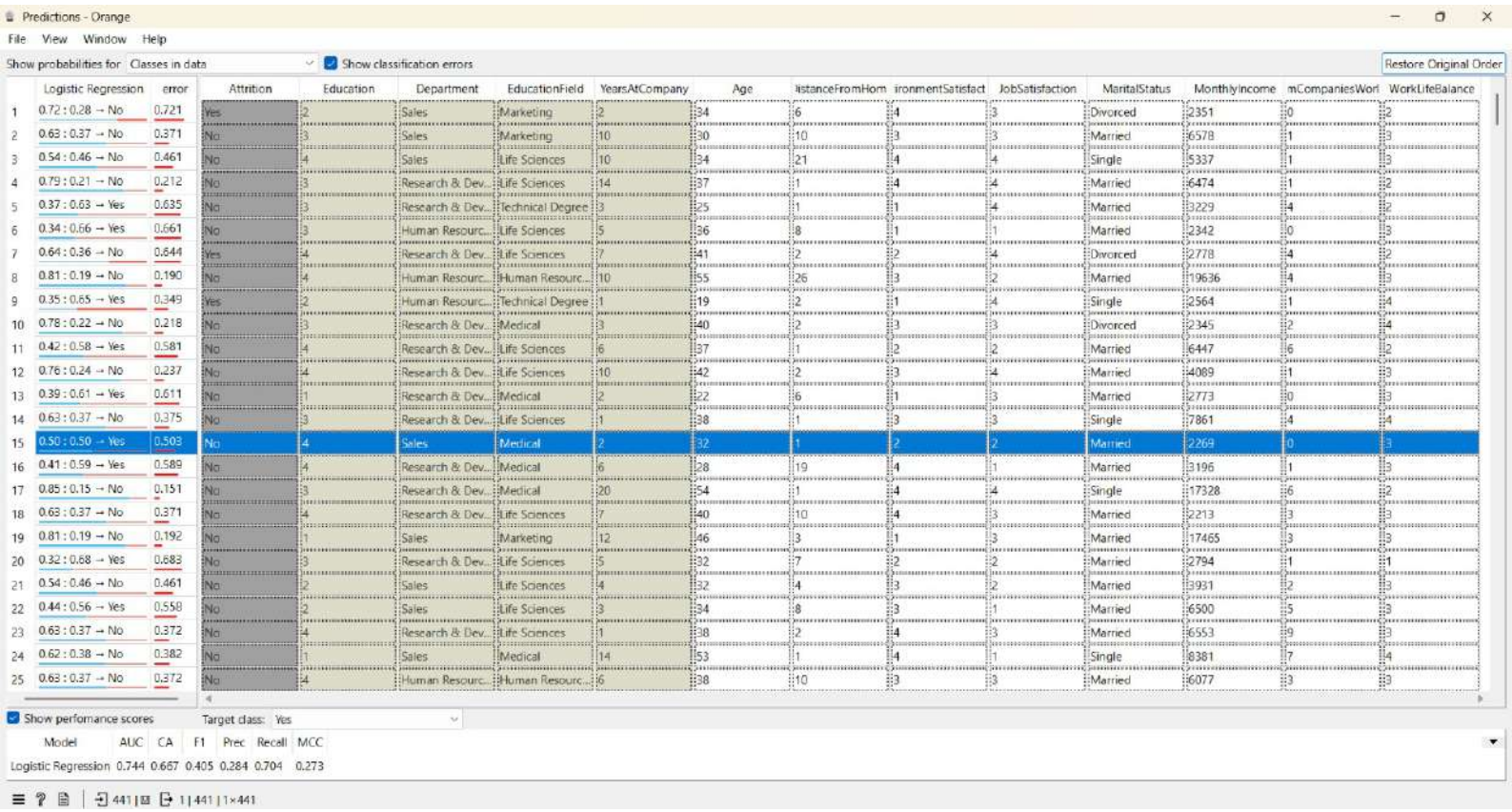

Capstone Project.ows - Orange
File Edit View Widget Window Options Help
Filter...
Data
File
CSV File Import
Datasets
SQL Table
Data Table
Paint Data
Data Info
Rank
Edit Domain
Color
Column Statistics
Save Data
Transform
Visualize
Model
Evaluate
Unsupervised
Logistic Regression
The logistic regression classification algorithm with LASSO (L1) or ridge (L2) regularization.
more...
CSV File Import
Data Table
Box Plot
Select Columns
Distributions
Rank
Data Sampler
Predictions
Confusion Matrix
ROC Analysis
Nomogram
Logistic Regression
Data Table (1)
Data
Remaining Data → Data
Evaluation Results
Model → Predictors
Model → Classifier
Data Sample → Data
Coefficients → Data

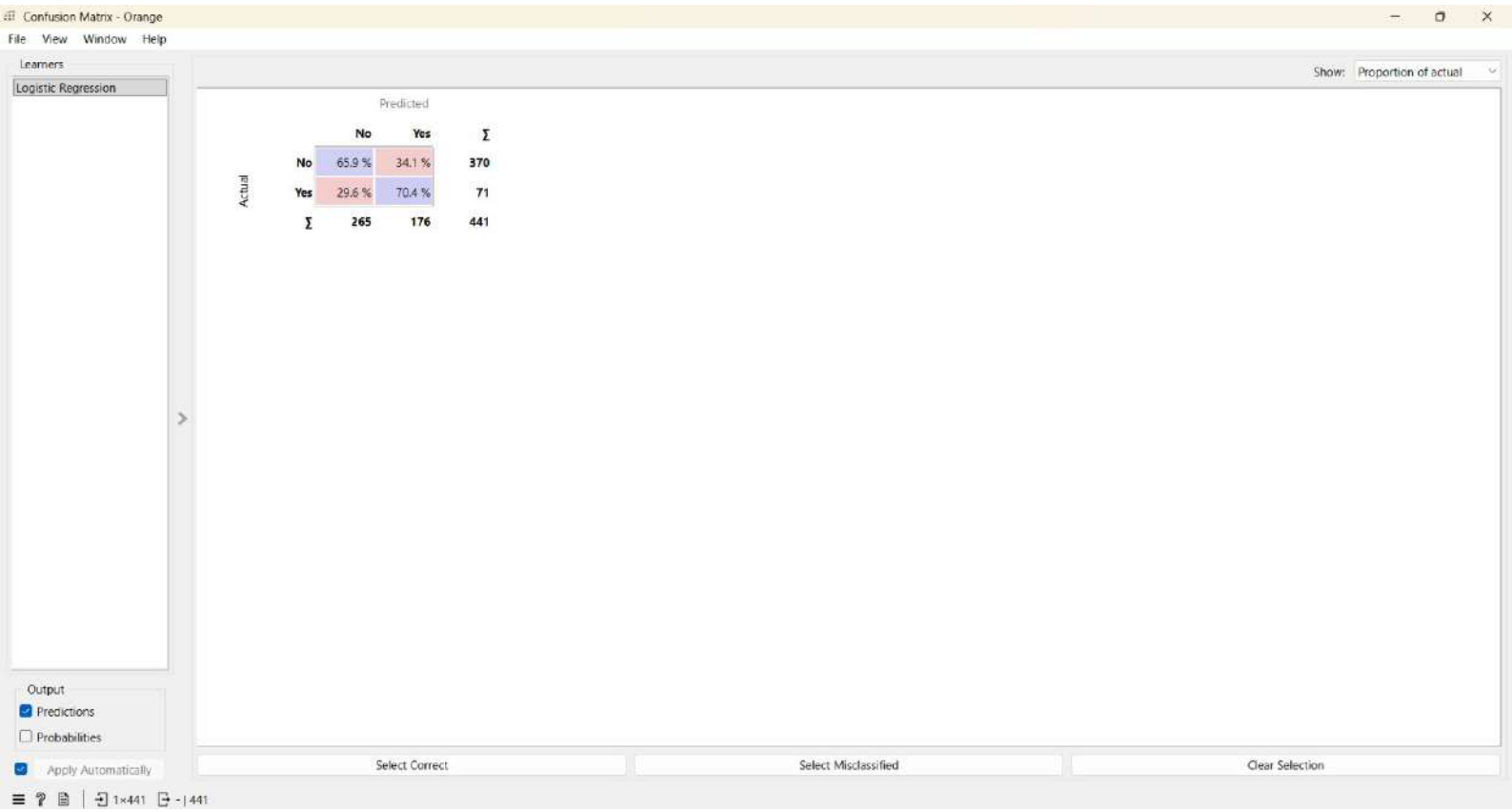

Select Columns - Orange
File View Window Help
Ignored
Filter
Features (8)
Filter
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
>
Target (1)
Attrition
Metas (4)
Education
Department
EducationField
YearsAtCompany
Reset
Ignore new variables by default
Send Automatically
1470 | - 1470 | 8

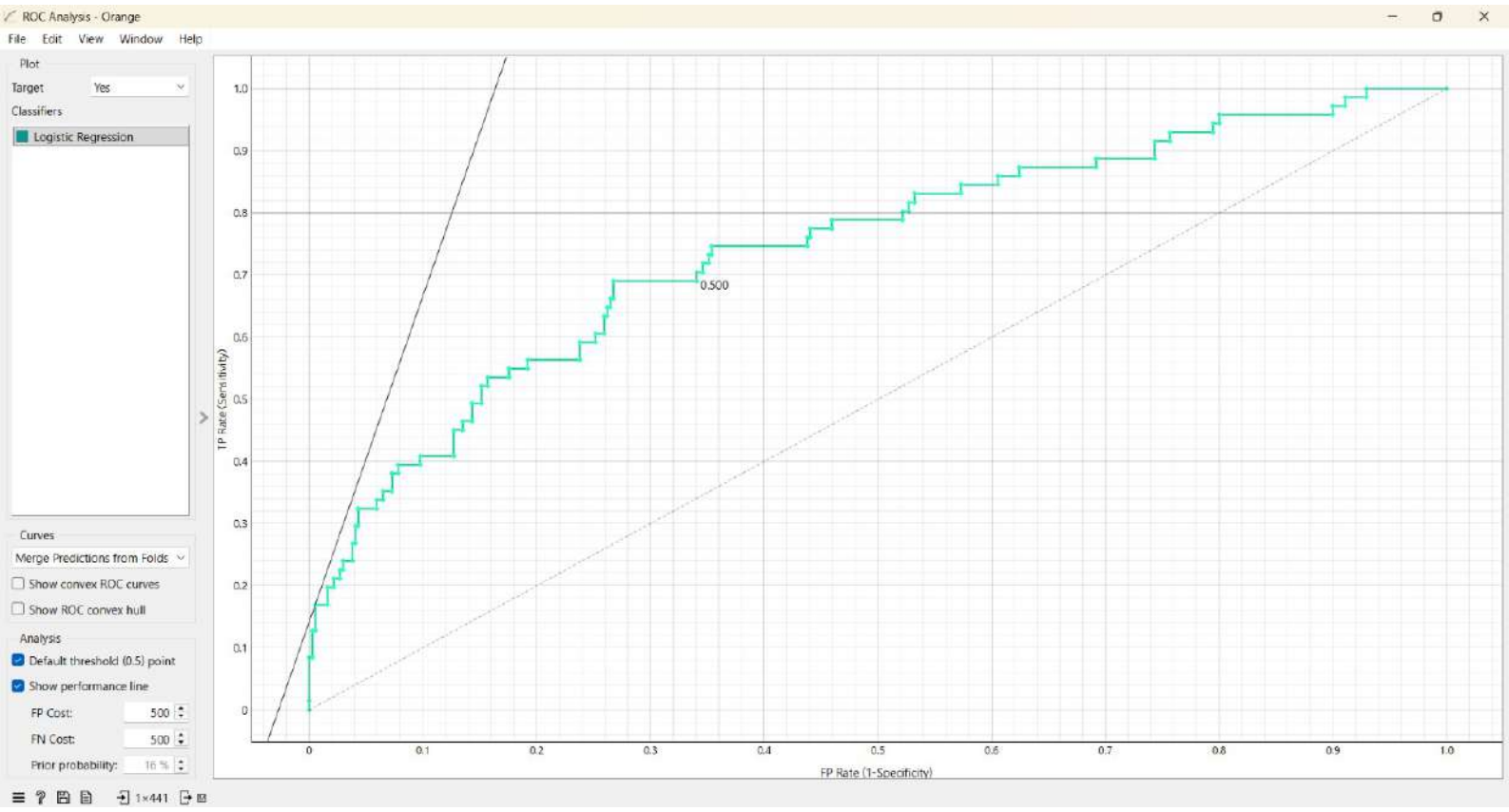

Capstone Project.ows - Orange
File Edit View Widget Window Options Help
Filter...
Data
File
CSV File Import
Datasets
SQL Table
Data Table
Paint Data
Data Info
Rank
Edit Domain
Color
Column Statistics
Save Data
Transform
Visualize
Model
Evaluate
Unsupervised
Data Sampler
Randomly draw a subset of data points from the input dataset.
more...
CSV File Import
Data Table
Box Plot
Select Columns
Distributions
Rank
Data Sampler
Logistic Regression
Data Table (1)
Confusion Matrix
ROC Analysis
Nomogram
Data Sampler - Orange
File View Window Help
Sampling Type
Fixed proportion of data:
70 %
Fixed sample size
Instances: 1
Sample with replacement
Cross validation
Number of subsets: 10
Unused subset: 1
Bootstrap
Options
Replicable (deterministic) sampling
Stratify sample (when possible)
Sample Data
1470
1029 | 441
Data
Remaining Data
Data Sample
Model
Coefficients
Data

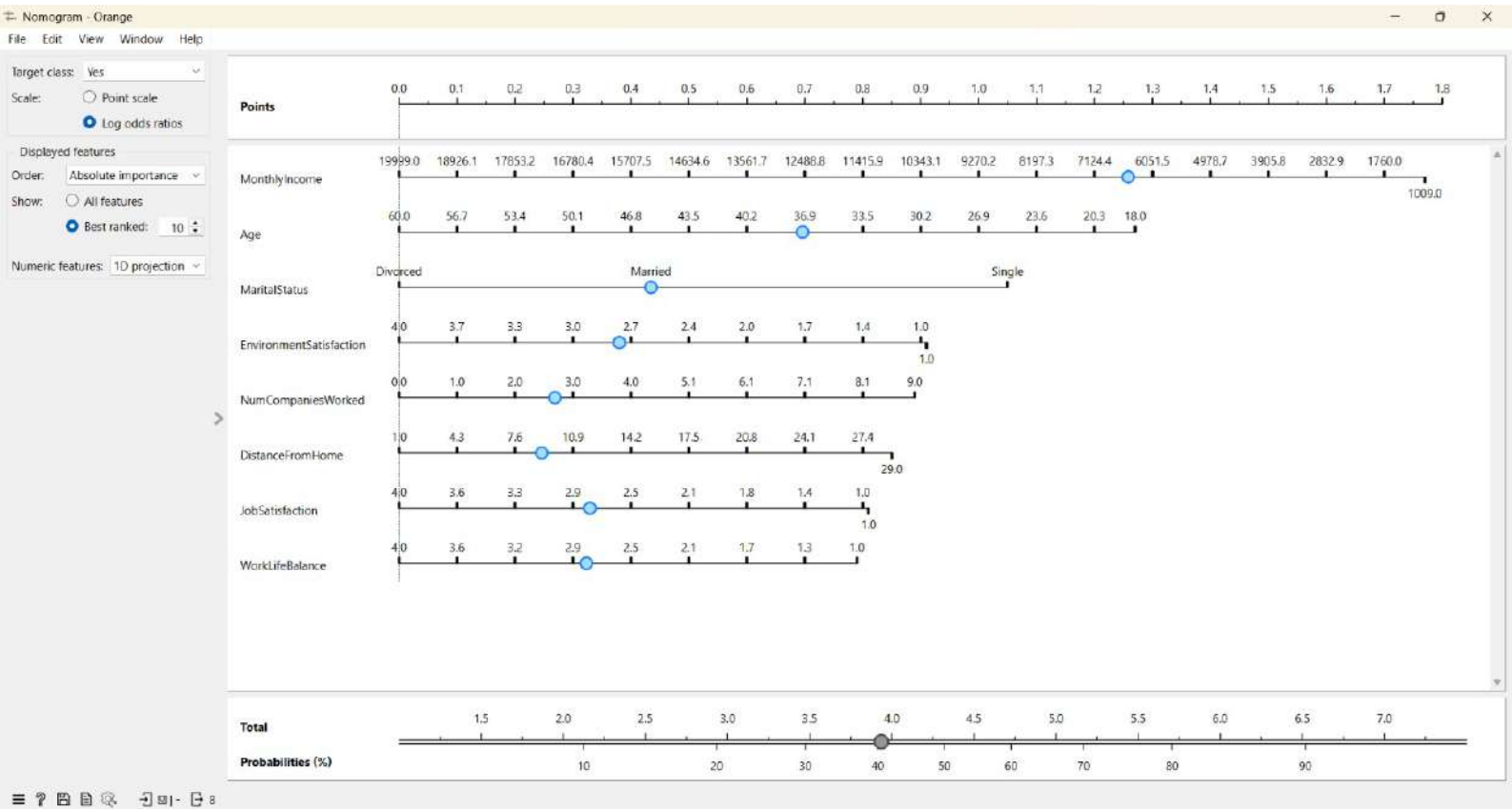

Capstone Project.ows - Orange
File Edit View Widget Window Options Help
Filter...
Data
File
CSV File Import
Datasets
SQL Table
Data Table
Paint Data
Data Info
Rank
Edit Domain
Color
Column Statistics
Save Data
Transform
Visualize
Model
Evaluate
Unsupervised
Logistic Regression
The logistic regression classification algorithm with LASSO (L1) or ridge (L2) regularization.
more...
CSV File Import
Data Table
Select Columns
Data Sampler
Logistic Regression - Orange
File View Window Help
Name
Logistic Regression
Regularization type: Ridge (L2)
Strength:
Weak
Strong
C=1
Balance class distribution
Apply Automatically
102
11
Predictions
Confusion Matrix
ROC Analysis
Nomogram
Logistic Regression
Data Table (1)
Data
Evaluation Results
Model → Predictions
Model → Classifier
Coefficients → Data
Data Sample → Data

Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


Predictions - Orange
File View Window Help
Show probabilities for Classes in data
Show classification errors
Restore Original Order
Logistic Regression error
Attrition Education Department EducationField YearsAtCompany Age listanceFromHom ironmentSatisfact JobSatisfaction MaritalStatus MonthlyIncome mCompaniesWorl WorkLifeBalance
1 0.72 : 0.28 → No 0.721 Yes 2 Sales Marketing 2 34 6 4 3 Divorced 2351 0 2
2 0.63 : 0.37 → No 0.371 No 3 Sales Marketing 10 30 10 3 3 Married 6578 1 3
3 0.54 : 0.46 → No 0.461 No 4 Sales Life Sciences 10 34 21 4 4 Single 5337 1 3
4 0.79 : 0.21 → No 0.212 No 3 Research & Dev... Life Sciences 14 37 1 4 4 Married 6474 1 2
5 0.37 : 0.63 → Yes 0.635 No 3 Research & Dev... Technical Degree 3 25 1 1 4 Married 3229 4 2
6 0.34 : 0.66 → Yes 0.661 No 3 Human Resourc... Life Sciences 5 36 8 1 1 Married 2342 0 3
7 0.64 : 0.36 → No 0.644 Yes 4 Research & Dev... Life Sciences 7 41 2 2 4 Divorced 2778 4 2
8 0.81 : 0.19 → No 0.190 No 4 Human Resourc... Human Resourc... 10 55 26 3 2 Married 19636 4 3
9 0.35 : 0.65 → Yes 0.349 Yes 2 Human Resourc... Technical Degree 1 19 2 1 4 Single 2564 1 4
10 0.78 : 0.22 → No 0.218 No 3 Research & Dev... Medical 3 40 2 3 3 Divorced 2345 2 4
11 0.42 : 0.58 → Yes 0.581 No 4 Research & Dev... Life Sciences 6 37 1 2 2 Married 6447 6 2
12 0.76 : 0.24 → No 0.237 No 4 Research & Dev... Life Sciences 10 42 2 3 4 Married 4089 1 3
13 0.39 : 0.61 → Yes 0.611 No 1 Research & Dev... Medical 2 22 6 1 3 Married 2773 0 3
14 0.63 : 0.37 → No 0.375 No 3 Research & Dev... Life Sciences 1 38 1 3 3 Single 7861 4 4
15 0.50 : 0.50 → Yes 0.503 No 4 Sales Medical 2 32 1 2 2 Married 2269 0 3
16 0.41 : 0.59 → Yes 0.589 No 4 Research & Dev... Medical 6 28 19 4 1 Married 3196 1 3
17 0.85 : 0.15 → No 0.151 No 3 Research & Dev... Medical 20 54 1 4 4 Single 17328 6 2
18 0.63 : 0.37 → No 0.371 No 4 Research & Dev... Life Sciences 7 40 10 4 3 Married 2213 3 3
19 0.81 : 0.19 → No 0.192 No 1 Sales Marketing 12 46 3 1 3 Married 17465 3 3
20 0.32 : 0.68 → Yes 0.683 No 3 Research & Dev... Life Sciences 5 32 7 2 2 Married 2794 1 1
21 0.54 : 0.46 → No 0.461 No 2 Sales Life Sciences 4 32 4 3 2 Married 3931 2 3
22 0.44 : 0.56 → Yes 0.558 No 2 Sales Life Sciences 3 34 8 3 1 Married 6500 5 3
23 0.63 : 0.37 → No 0.372 No 4 Research & Dev... Life Sciences 1 38 2 4 3 Married 6553 9 3
24 0.62 : 0.38 → No 0.382 No 1 Sales Medical 14 53 1 4 1 Single 8381 7 4
25 0.63 : 0.37 → No 0.372 No 4 Human Resourc... Human Resourc... 6 38 10 3 3 Married 6077 3 3
Show performance scores
Target class: Yes
Model AUC CA F1 Prec Recall MCC
Logistic Regression 0.744 0.667 0.405 0.284 0.704 0.273
441 | 441 | 1×441


Confusion Matrix - Orange
File View Window Help
Learners
Logistic Regression
Show: Proportion of actual
Predicted
No Yes ∑
Actual
No 65.9 % 34.1 % 370
Yes 29.6 % 70.4 % 71
∑ 265 176 441
Output
Predictions
Probabilities
Apply Automatically
Select Correct
Select Misclassified
Clear Selection
1×441
- | 441


ROC Analysis - Orange
File Edit View Window Help
Plot
Target Yes
Classifiers
Logistic Regression
Curves
Merge Predictions from Folds
Show convex ROC curves
Show ROC convex hull
Analysis
Default threshold (0.5) point
Show performance line
FP Cost: 500
FN Cost: 500
Prior probability: 16 %
1×441
TP Rate (Sensitivity)
FP Rate (1-Specificity)
0.500
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0


Nomogram - Orange
File Edit View Window Help
Target class: Yes
Scale: Point scale
Log odds ratios
Displayed features
Order: Absolute importance
Show: All features
Best ranked: 10
Numeric features: 1D projection
Points
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8
MonthlyIncome
19999.0 18926.1 17853.2 16780.4 15707.5 14634.6 13561.7 12488.8 11415.9 10343.1 9270.2 8197.3 7124.4 6051.5 4978.7 3905.8 2832.9 1760.0 1009.0
Age
60.0 56.7 53.4 50.1 46.8 43.5 40.2 36.9 33.5 30.2 26.9 23.6 20.3 18.0
MaritalStatus
Divorced Married Single
EnvironmentSatisfaction
4.0 3.7 3.3 3.0 2.7 2.4 2.0 1.7 1.4 1.0 1.0
NumCompaniesWorked
0.0 1.0 2.0 3.0 4.0 5.1 6.1 7.1 8.1 9.0
DistanceFromHome
1.0 4.3 7.6 10.9 14.2 17.5 20.8 24.1 27.4 29.0
JobSatisfaction
4.0 3.6 3.3 2.9 2.5 2.1 1.8 1.4 1.0 1.0
WorkLifeBalance
4.0 3.6 3.2 2.9 2.5 2.1 1.7 1.3 1.0
Total
1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5 6.0 6.5 7.0
Probabilities (%)
10 20 30 40 50 60 70 80 90

Data Table (1) - Orange

File Edit View Window Help

Info
11 instances (no missing data)
1 feature
No target variable.
1 meta attribute

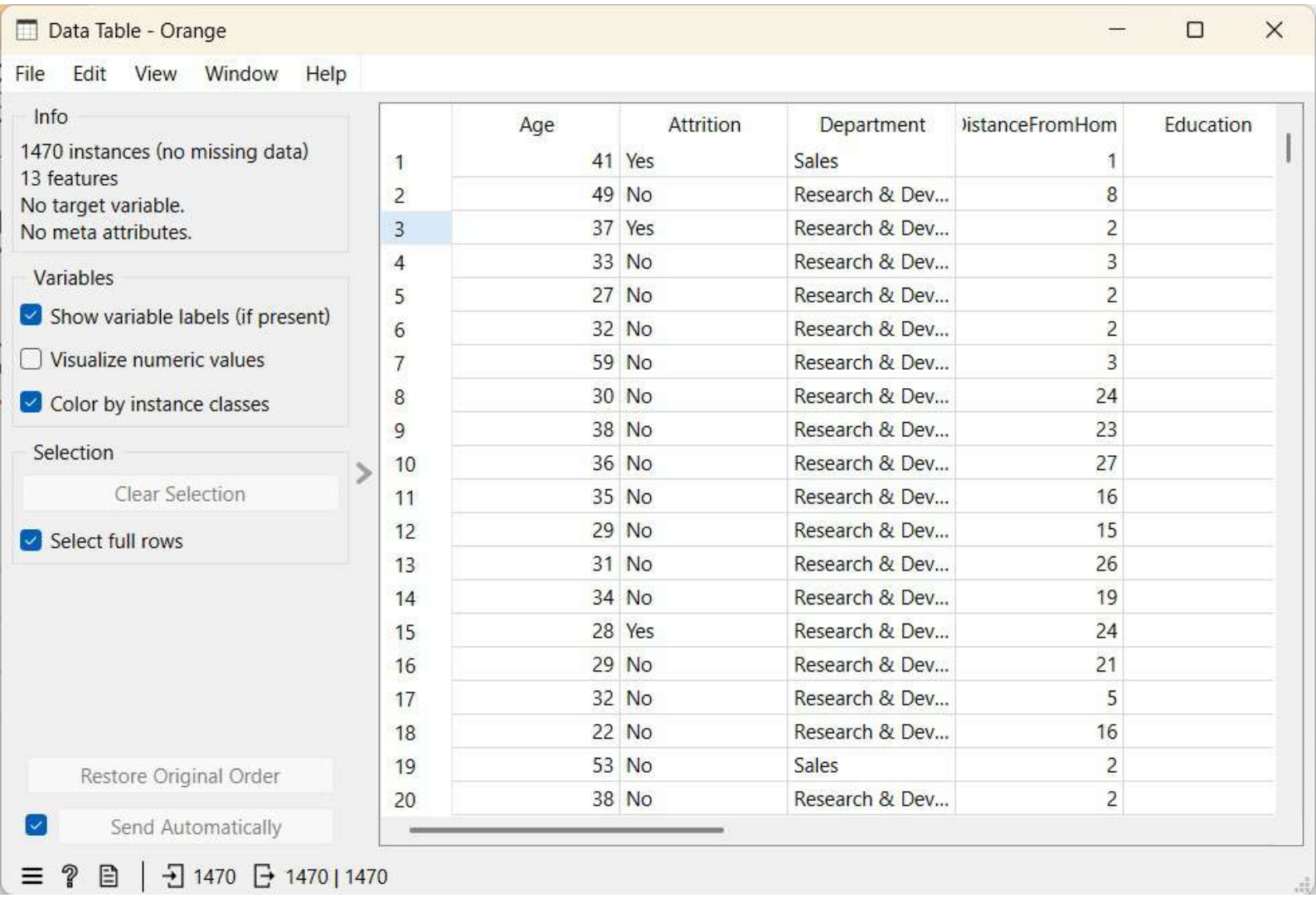
Variables
- [x] Show variable labels (if present)
- [ ] Visualize numeric values
- [x] Color by instance classes

Selection
Clear Selection
- [x] Select full rows

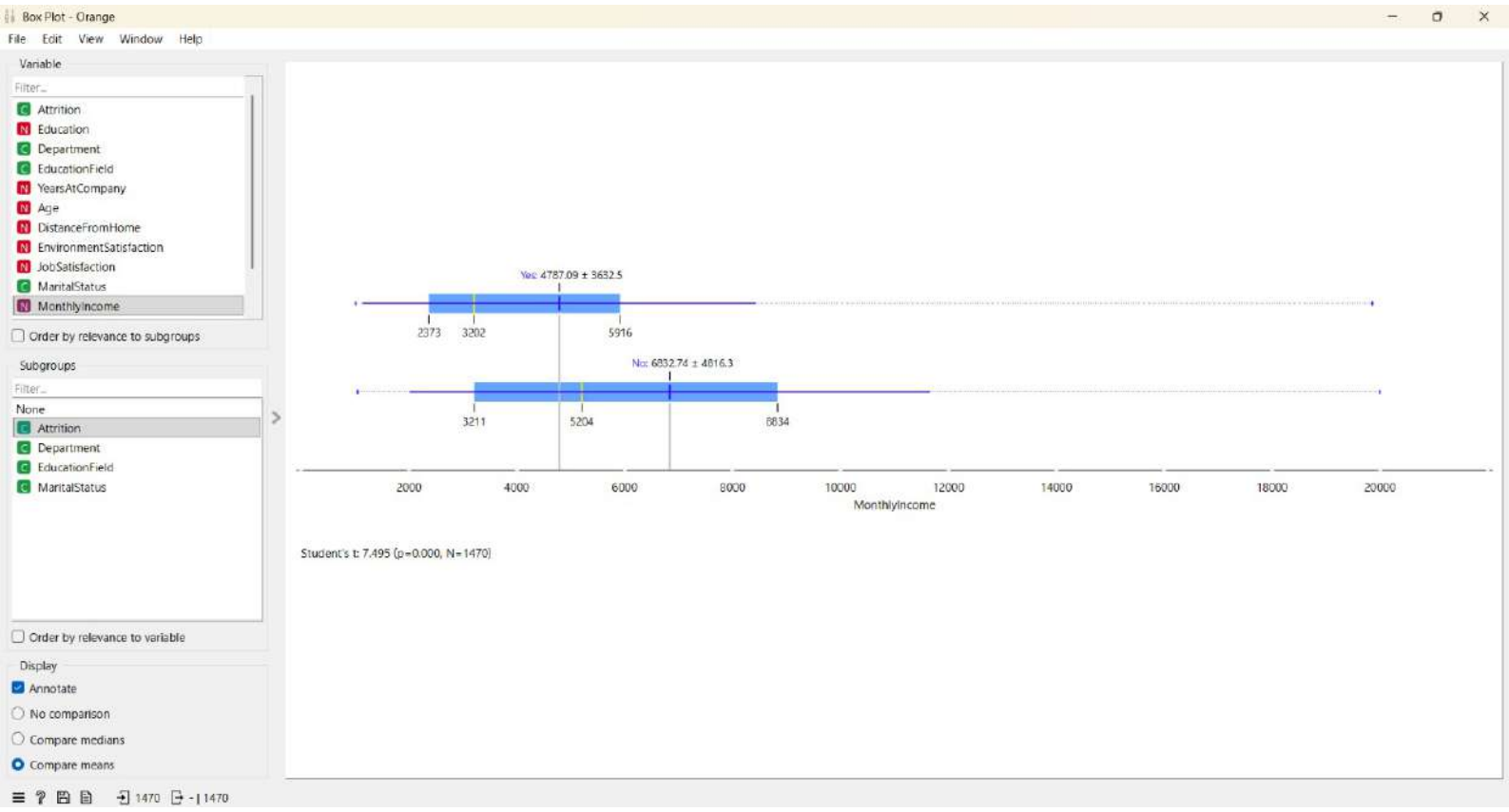
	name	Yes
1	intercept	3.16066
2	Age	-0.0302165
3	DistanceFromHome	0.030259
4	EnvironmentSatisfaction	-0.301099
5	JobSatisfaction	-0.269489
6	MaritalStatus=Divorced	-0.507018
7	MaritalStatus=Married	-0.072393
8	MaritalStatus=Single	0.542819
9	MonthlyIncome	-9.30487e-05
10	NumCompaniesWorked	0.0985023
11	WorkLifeBalance	-0.262115

Restore Original Order
- [x] Send Automatically

11 | 11 | 11



	Age	Attrition	Department	istanceFromHom	Education
1	41	Yes	Sales	1	
2	49	No	Research & Dev...	8	
3	37	Yes	Research & Dev...	2	
4	33	No	Research & Dev...	3	
5	27	No	Research & Dev...	2	
6	32	No	Research & Dev...	2	
7	59	No	Research & Dev...	3	
8	30	No	Research & Dev...	24	
9	38	No	Research & Dev...	23	
10	36	No	Research & Dev...	27	
11	35	No	Research & Dev...	16	
12	29	No	Research & Dev...	15	
13	31	No	Research & Dev...	26	
14	34	No	Research & Dev...	19	
15	28	Yes	Research & Dev...	24	
16	29	No	Research & Dev...	21	
17	32	No	Research & Dev...	5	
18	22	No	Research & Dev...	16	
19	53	No	Sales	2	
20	38	No	Research & Dev...	2	


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
1470
- | 1470
Yes: 4787.09 ± 3632.5
2373
3202
5916
No: 6832.74 ± 4816.3
3211
5204
6834
2000
4000
6000
8000
10000
12000
14000
16000
18000
20000
MonthlyIncome
Student's t: 7.495 (p=0.000, N=1470)

Box Plot - Orange

File Edit View Window Help

Variable

Filter...

- Attrition
- Education
- Department
- EducationField
- YearsAtCompany
- Age
- DistanceFromHome
- EnvironmentSatisfaction
- JobSatisfaction
- MaritalStatus
- MonthlyIncome

Order by relevance to subgroups

Subgroups

Filter...

None

- Attrition
- Department
- EducationField
- MaritalStatus

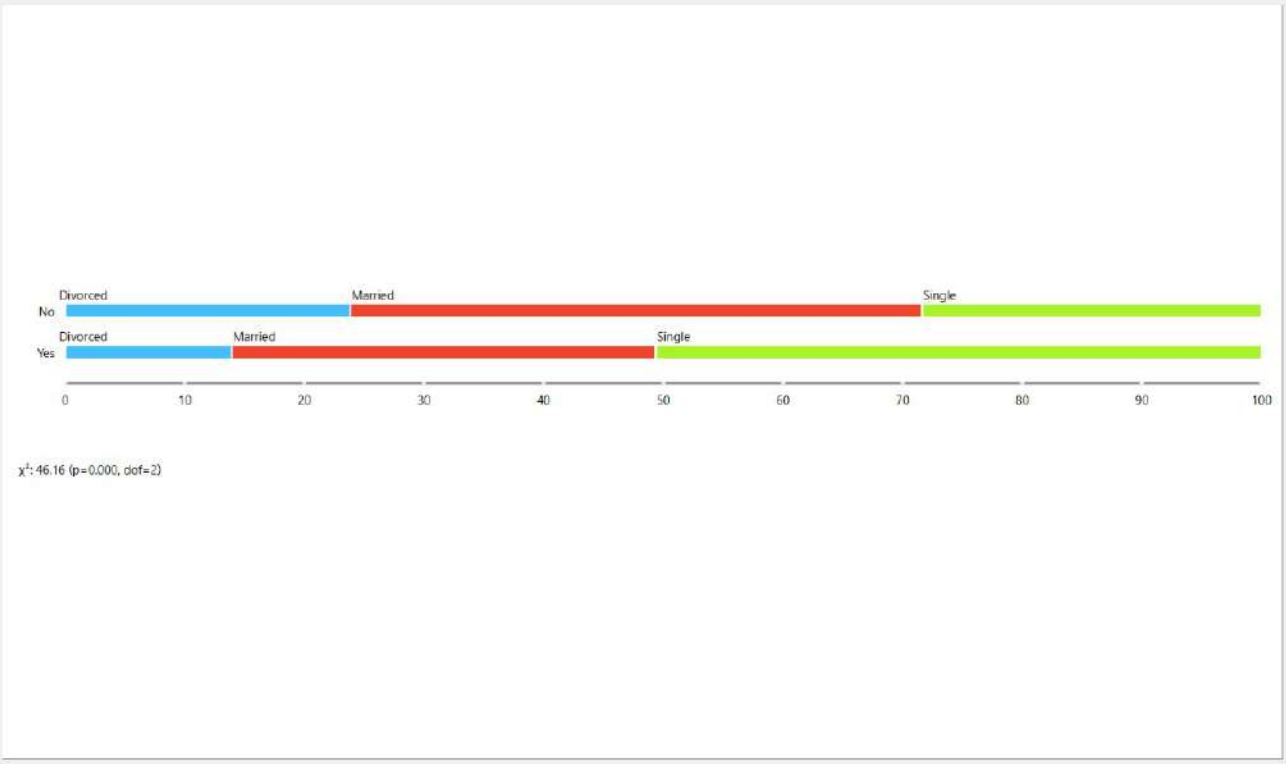
Order by relevance to variable

Display

Stretch bars

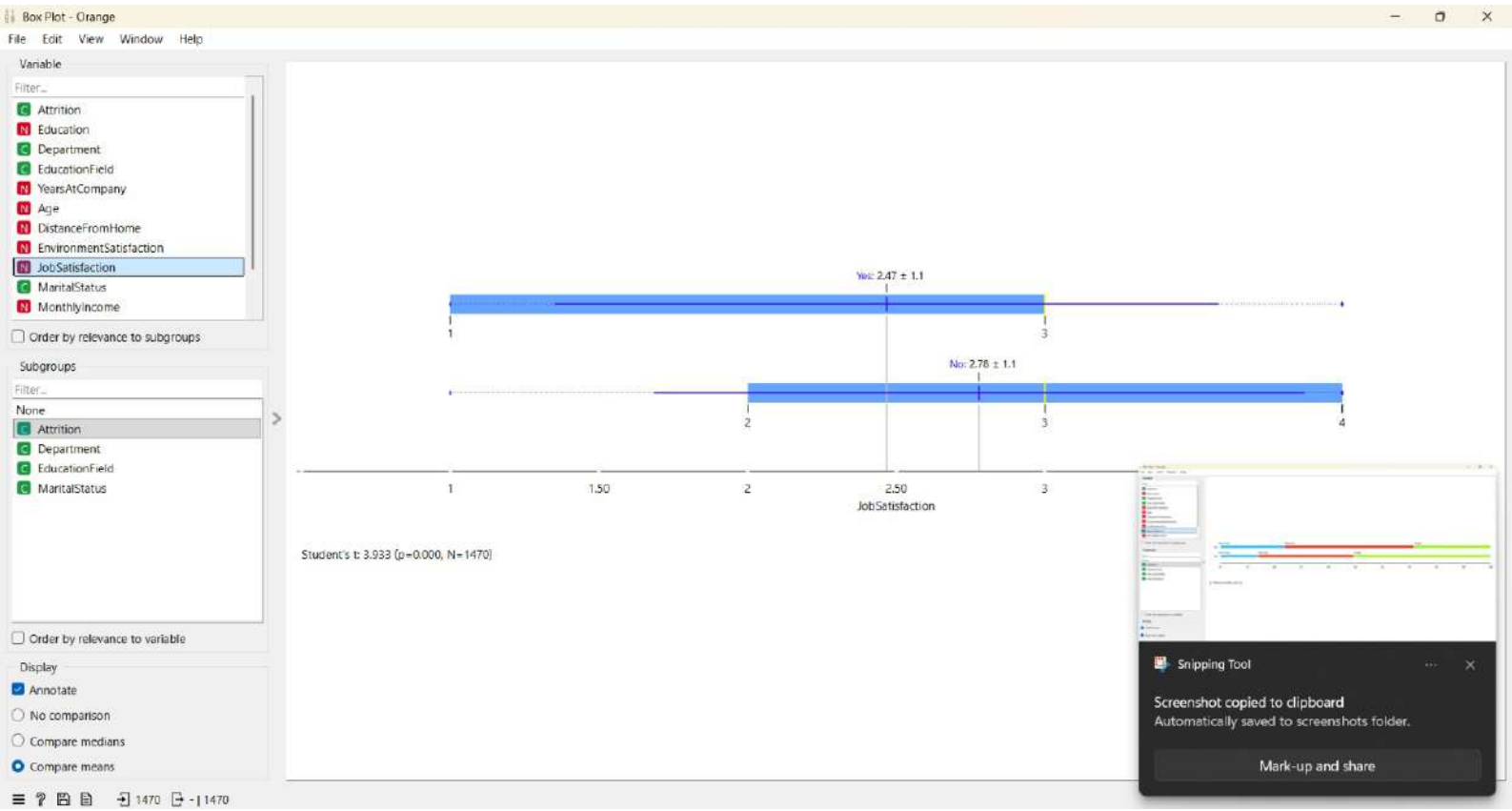
Show box labels

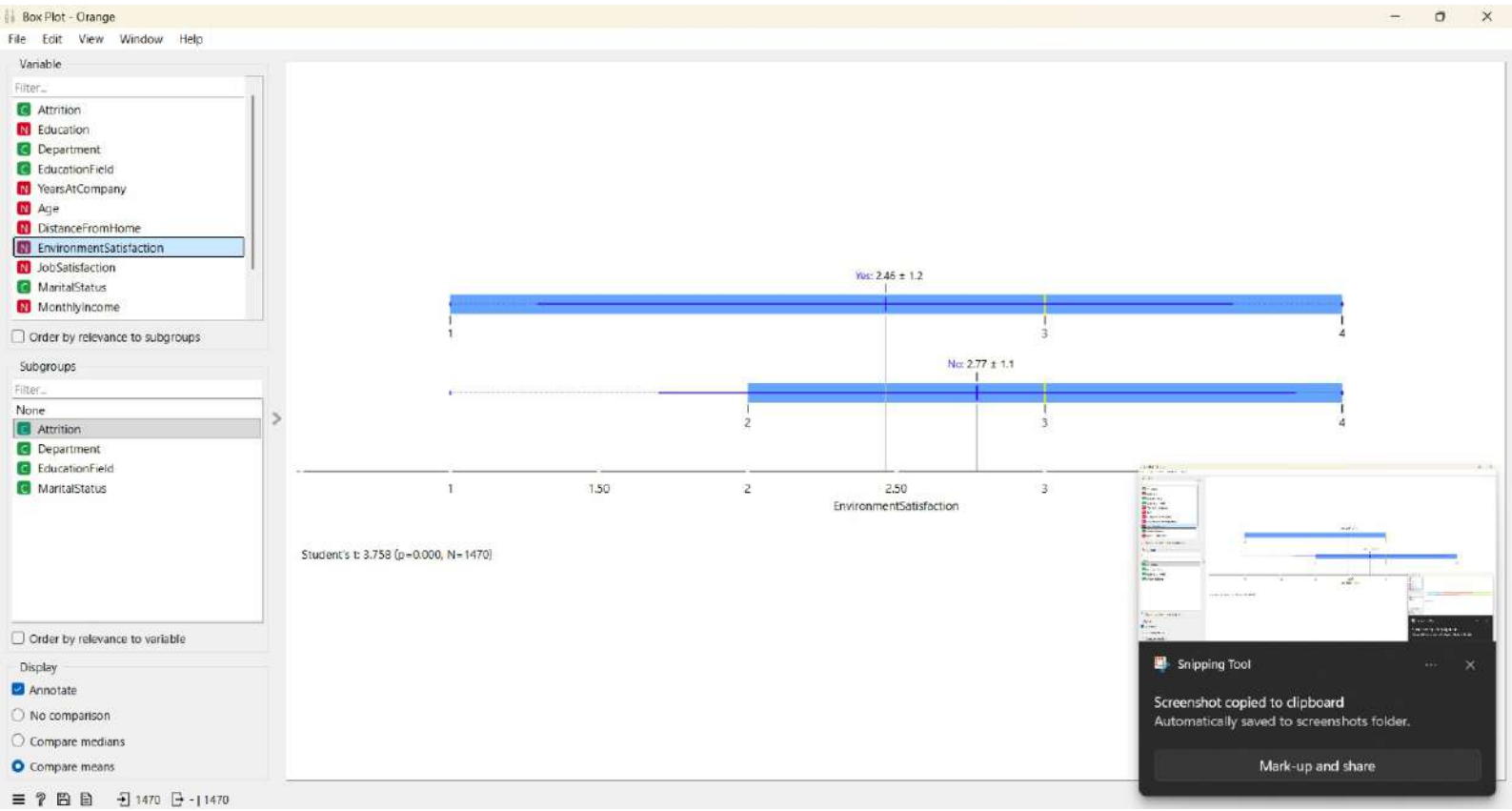
Sort by subgroup frequencies

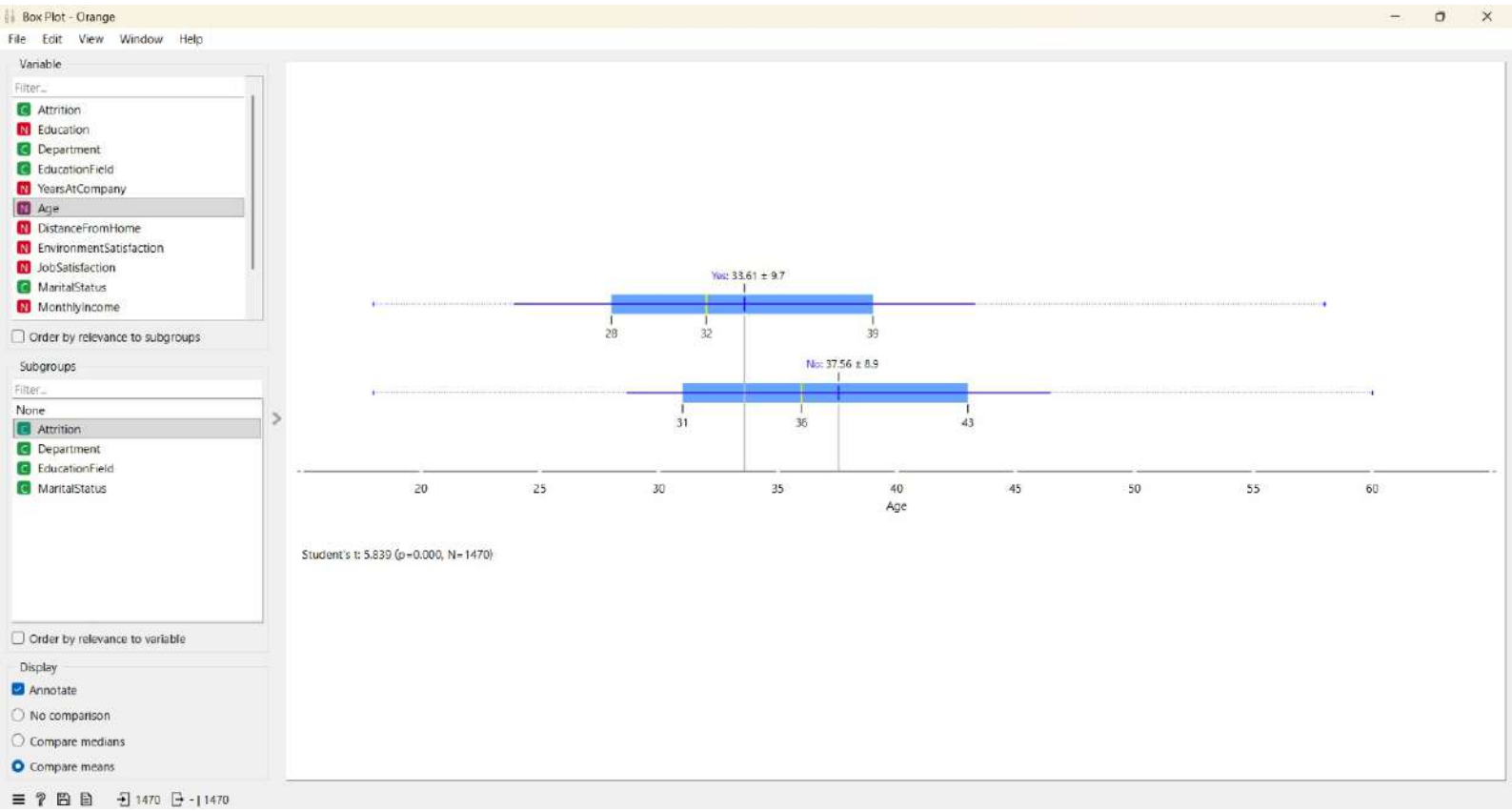


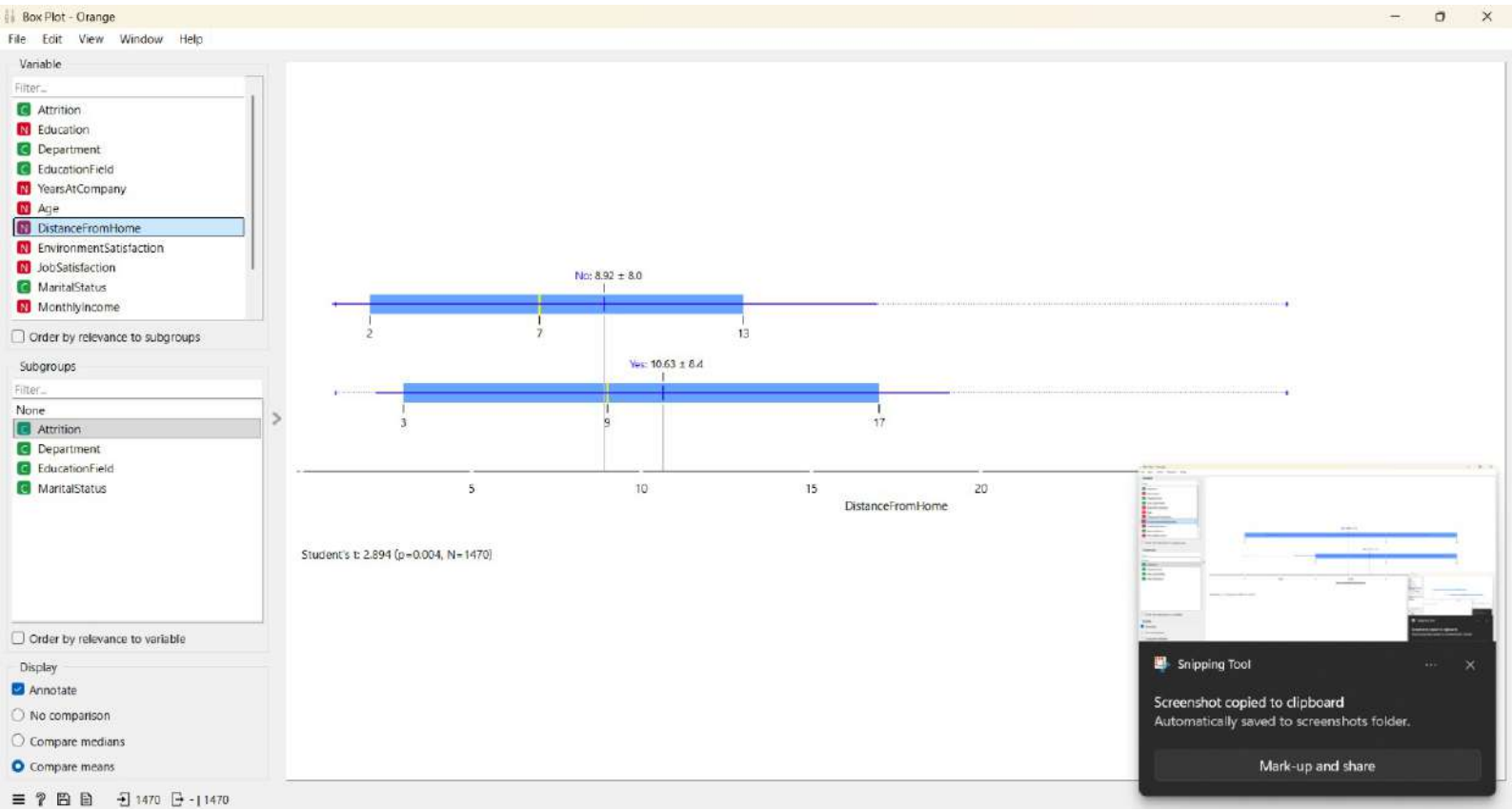
χ^2: 46.16 (p=0.000, dof=2)

1470 - | 1470


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
Yes: 2.47 ± 1.1
No: 2.78 ± 1.1
1
1.50
2
2.50
3
4
JobSatisfaction
Student's t: 3.933 (p=0.000, N=1470)
1470
1470
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
1470
- | 1470
Yes: 2.46 ± 1.2
No: 2.77 ± 1.1
1
1.50
2
2.50
3
4
EnvironmentSatisfaction
Student's t: 3.758 (p=0.000, N=1470)
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
1470
- | 1470
Yes: 33.61 ± 9.7
28
32
39
No: 37.56 ± 8.9
31
36
43
20
25
30
35
40
45
50
55
60
Age
Student's t: 5.839 (p=0.000, N=1470)


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
No: 8.92 ± 8.0
2
7
13
Yes: 10.63 ± 8.4
3
9
17
5
10
15
20
DistanceFromHome
Student's t: 2.894 (p=0.004, N=1470)
1470
1470
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share

Box Plot - Orange

File Edit View Window Help

Variable

Filter...

- Attrition
- Education
- Department
- EducationField
- YearsAtCompany
- Age
- DistanceFromHome
- EnvironmentSatisfaction
- JobSatisfaction
- MaritalStatus
- MonthlyIncome

Order by relevance to subgroups

Subgroups

Filter...

None

- Attrition
- Department
- EducationField
- MaritalStatus

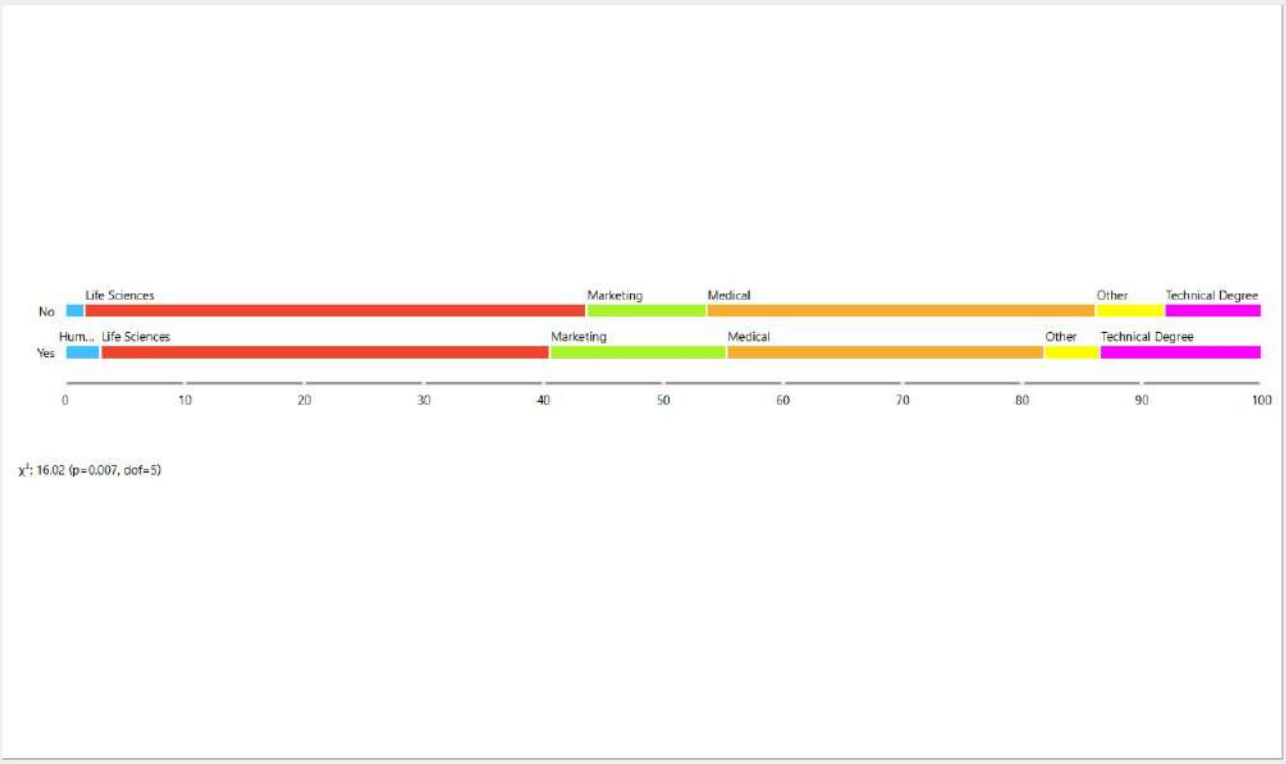
Order by relevance to variable

Display

Stretch bars

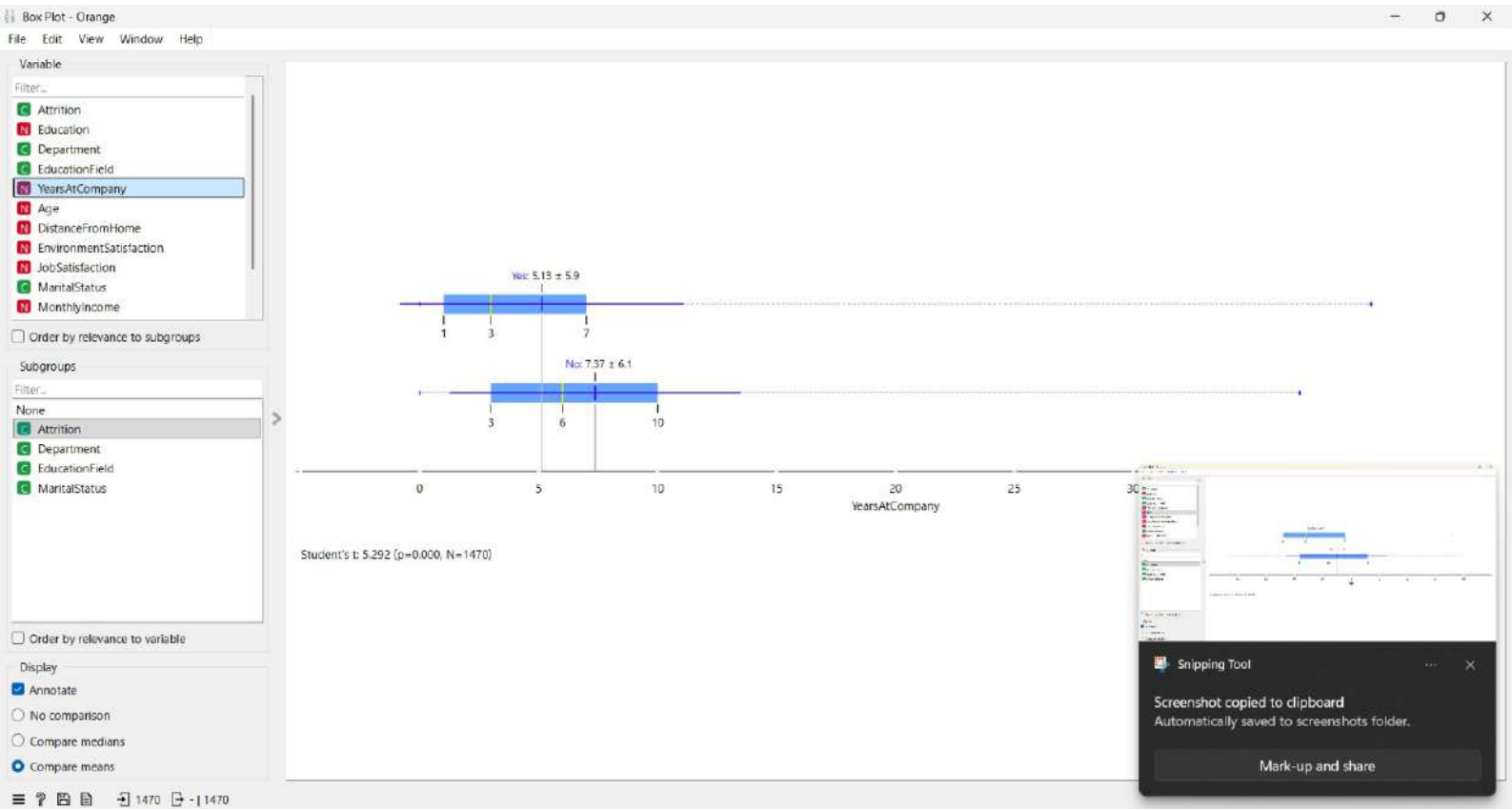
Show box labels

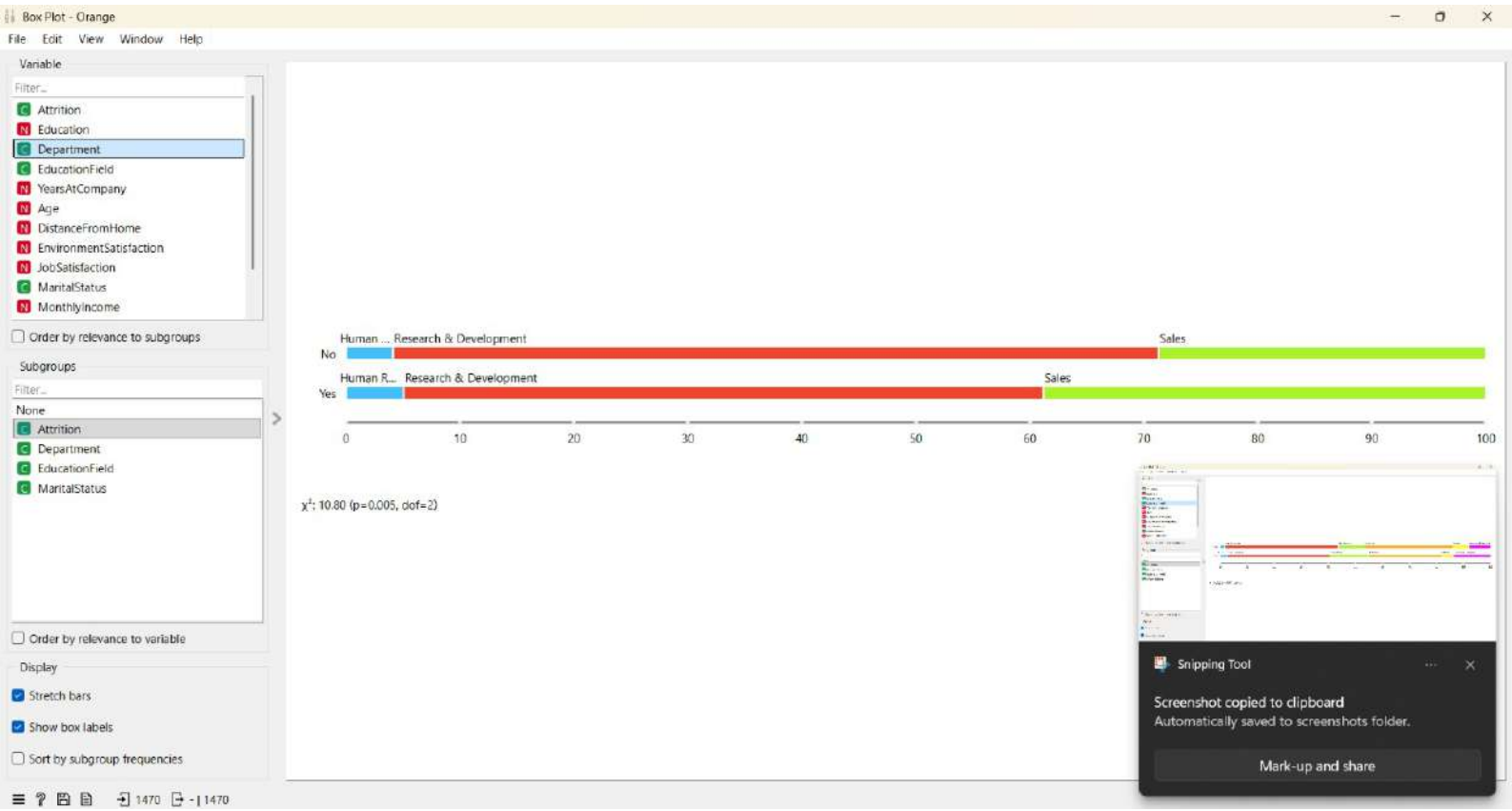
Sort by subgroup frequencies

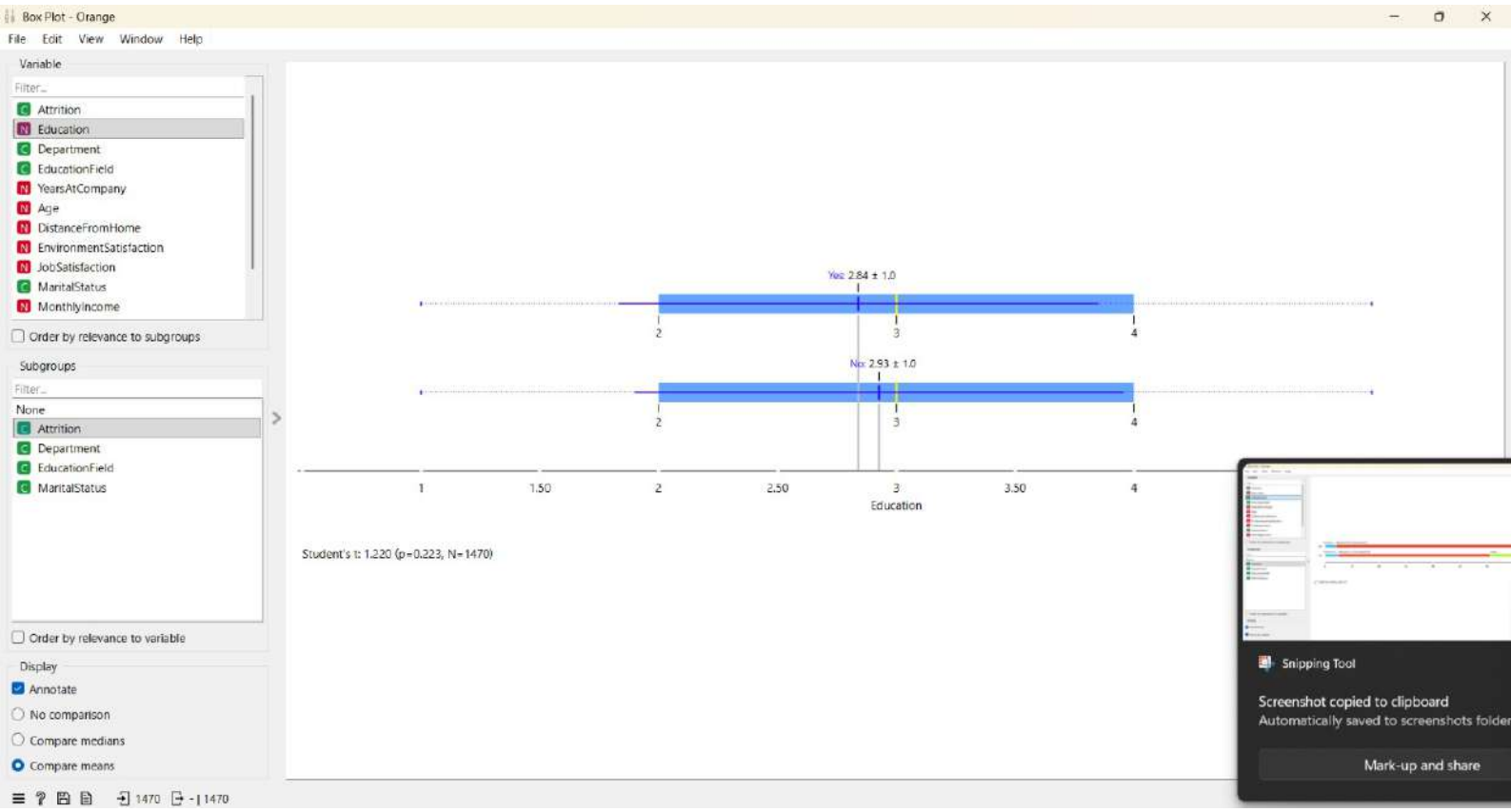


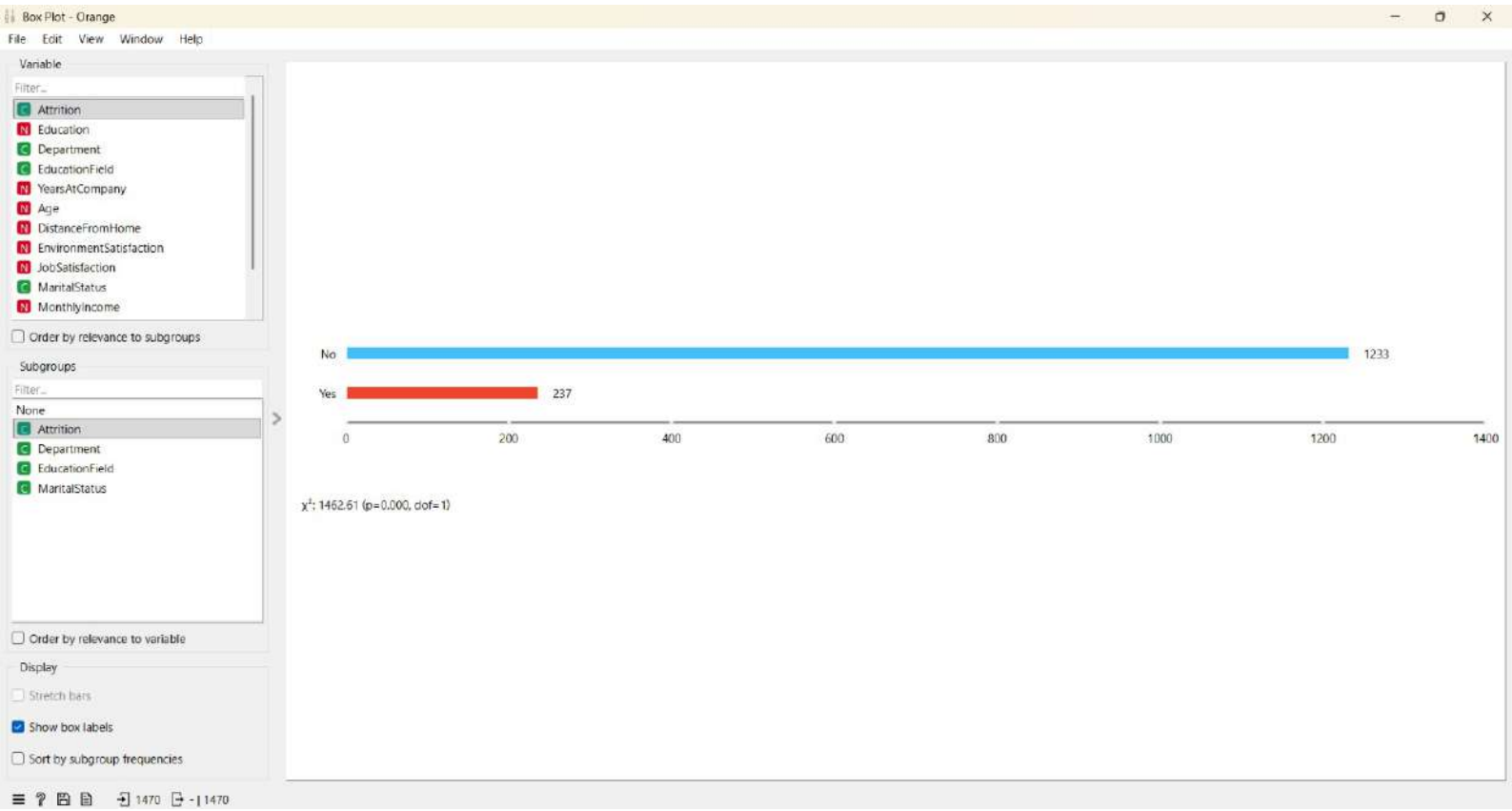
χ^2: 16.02 (p=0.007, dof=5)

1470 - | 1470


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
Yes: 5.13 ± 5.9
No: 7.37 ± 6.1
1
3
7
3
6
10
0
5
10
15
20
25
YearsAtCompany
Student's t: 5.292 (p=0.000, N=1470)
1470
1470
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Stretch bars
Show box labels
Sort by subgroup frequencies
No
Yes
Human ...
Research & Development
Sales
Human R...
0 10 20 30 40 50 60 70 80 90 100
χ²: 10.80 (p=0.005, dof=2)
1470
1470
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
1470
- | 1470
Yes: 2.84 ± 1.0
No: 2.93 ± 1.0
2
3
4
1
1.50
2
2.50
3
3.50
4
Education
Student's t: 1.220 (p=0.223, N=1470)
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder
Mark-up and share


Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Stretch bars
Show box labels
Sort by subgroup frequencies
No
Yes
1233
237
0
200
400
600
800
1000
1200
1400
χ²: 1462.61 (p=0.000, dof=1)
1470
1470

Box Plot - Orange

File Edit View Window Help

Variable

Filter...

- Department
- EducationField
- YearsAtCompany
- Age
- DistanceFromHome
- EnvironmentSatisfaction
- JobSatisfaction
- MaritalStatus
- MonthlyIncome
- NumCompaniesWorked
- WorkLifeBalance

Order by relevance to subgroups

Subgroups

Filter...

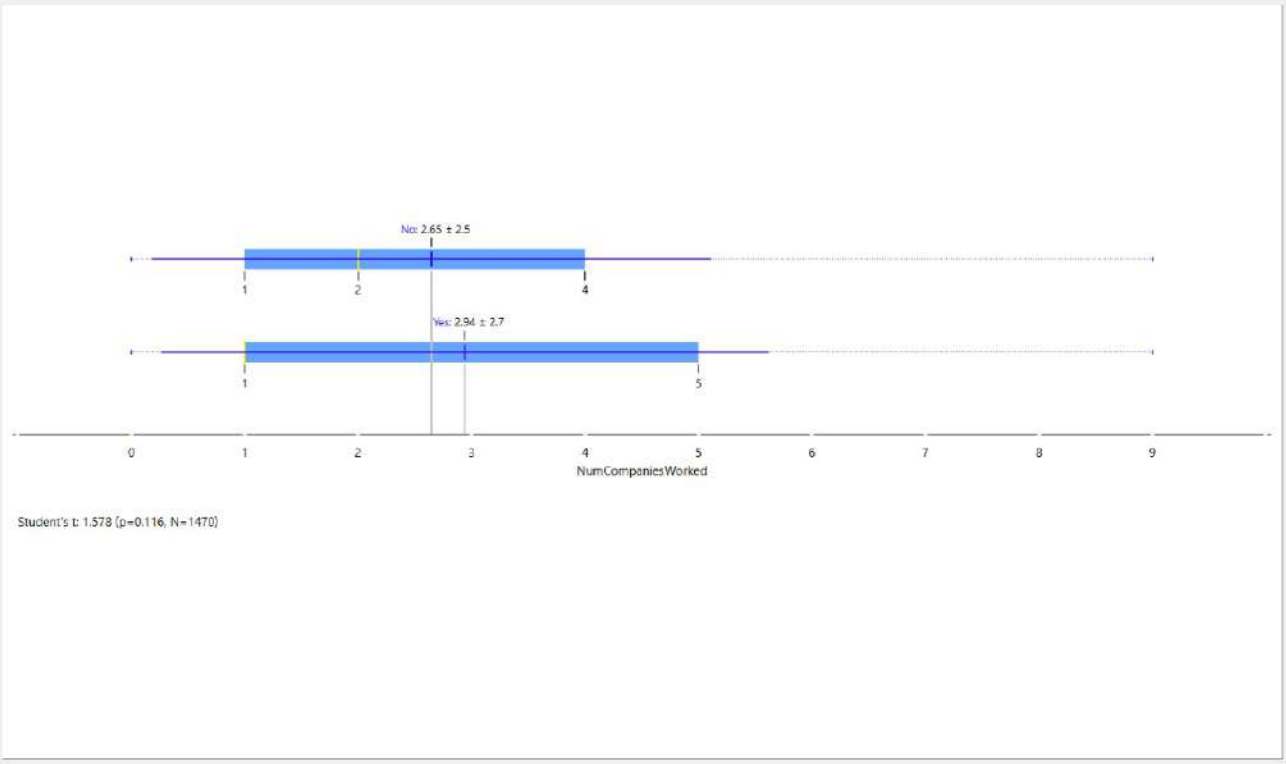
None

- Attrition
- Department
- EducationField
- MaritalStatus

Order by relevance to variable

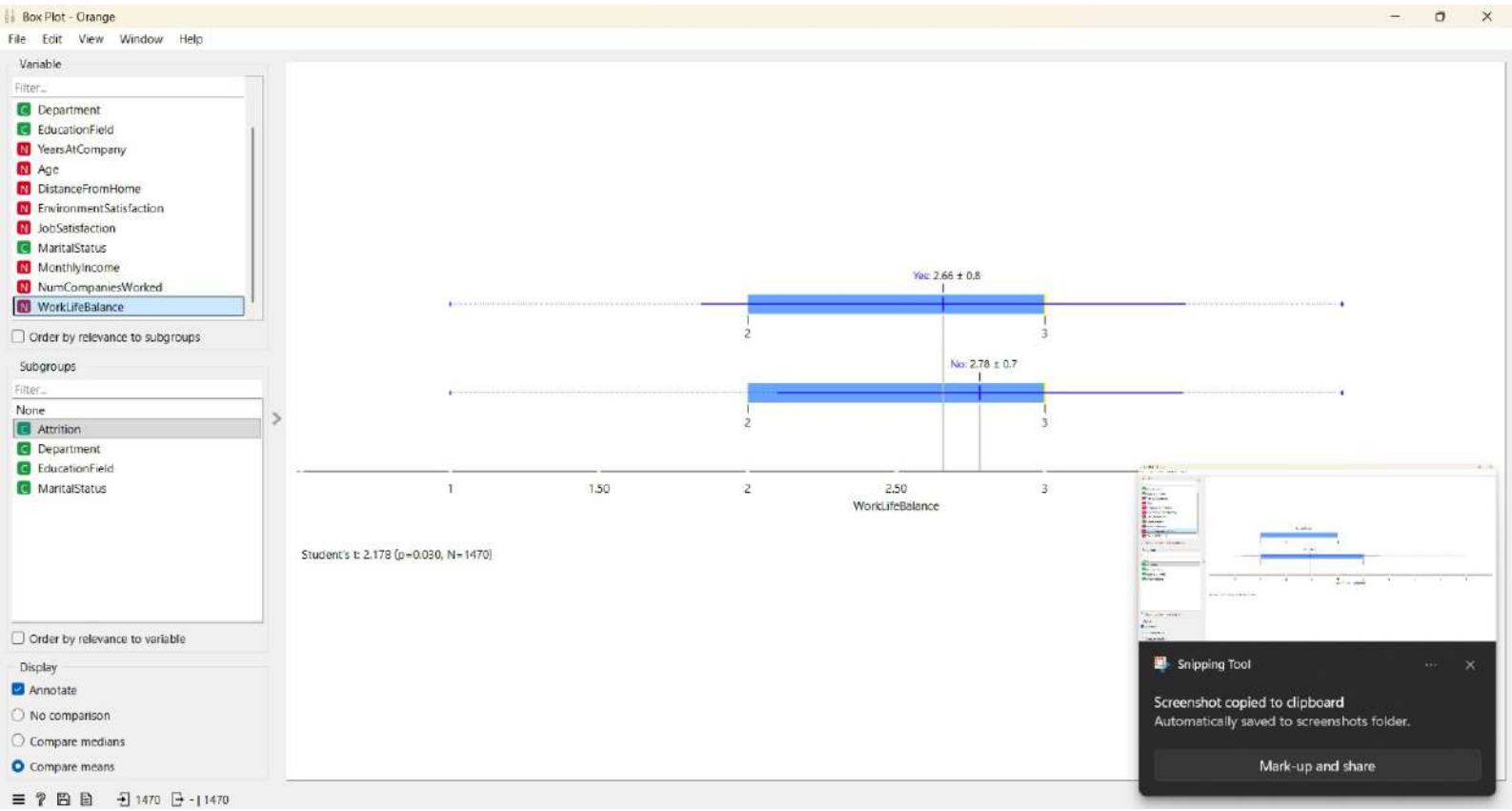
Display

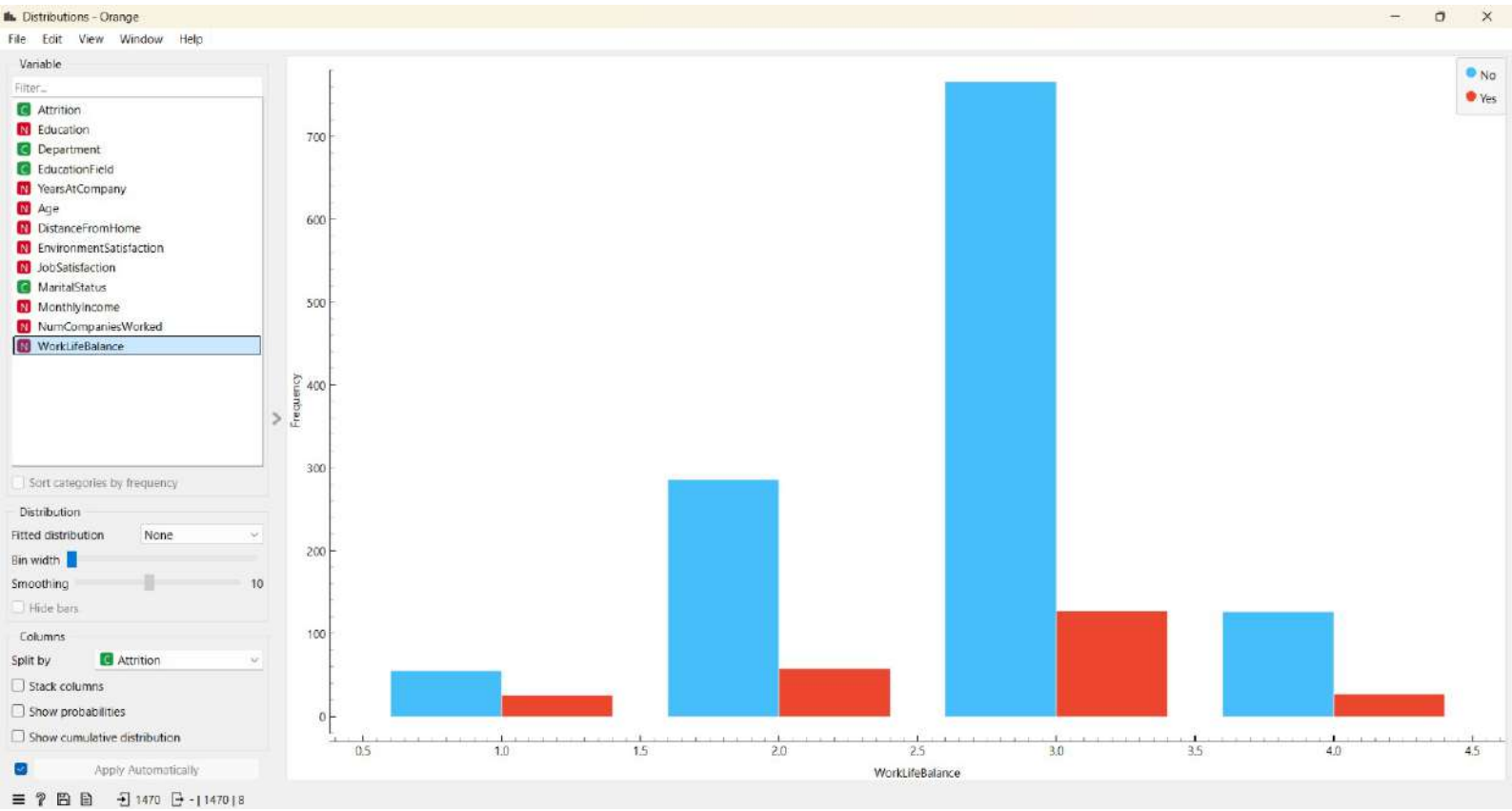
- Annotate
- No comparison
- Compare medians
- Compare means

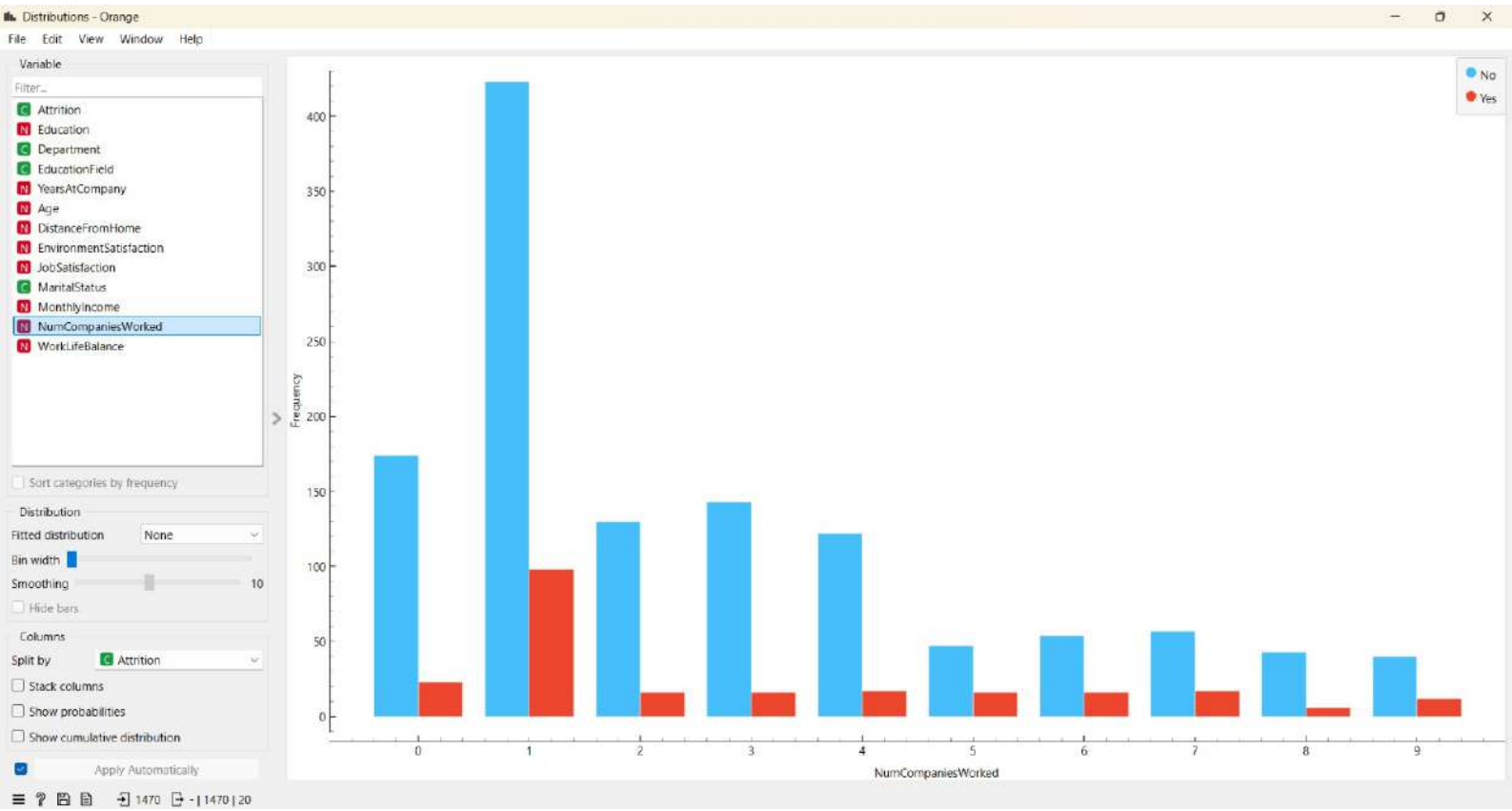


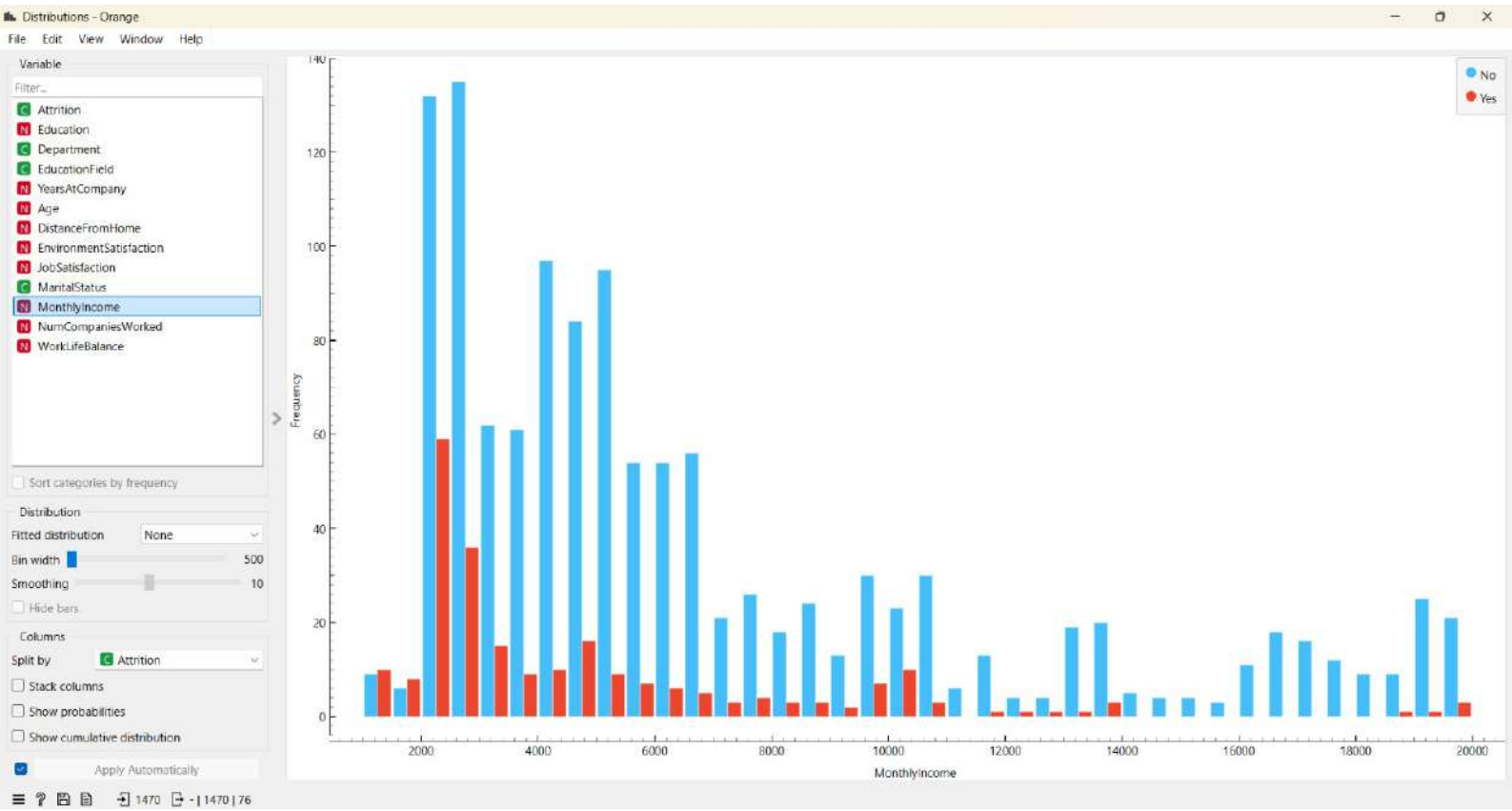
Student's t: 1.578 (p=0.116, N=1470)

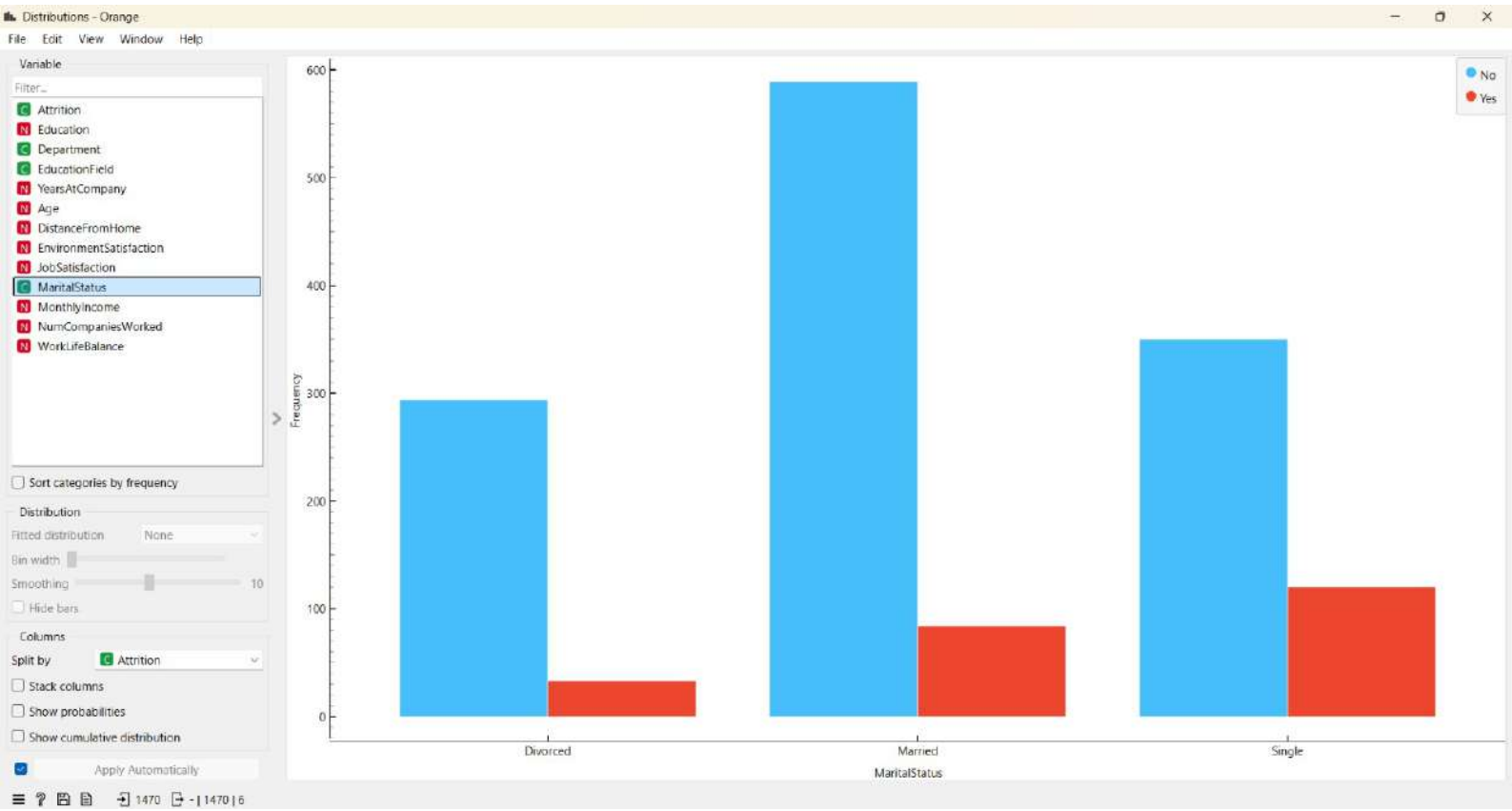
1470 - | 1470

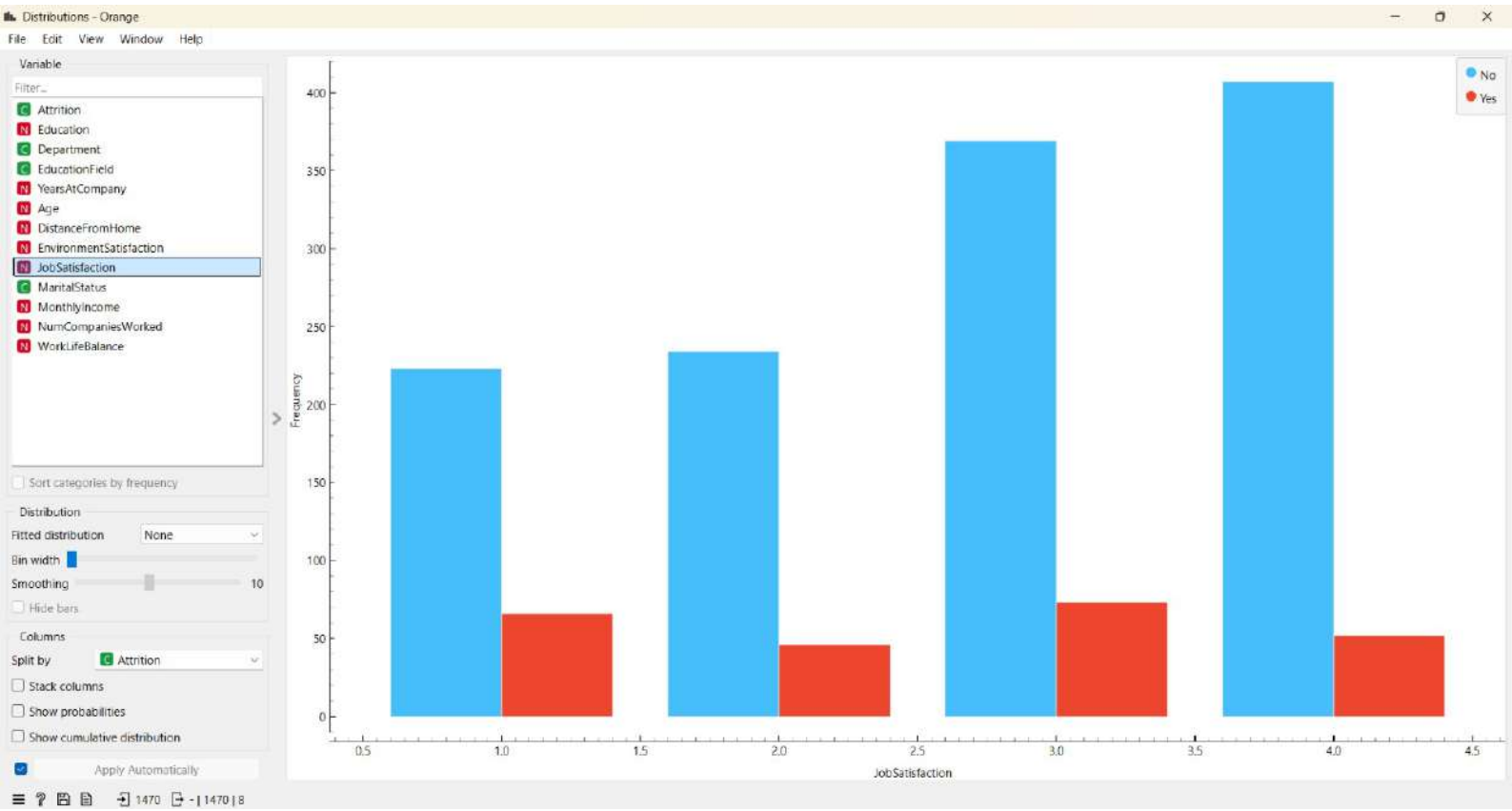

Box Plot - Orange
File Edit View Window Help
Variable
Filter...
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Order by relevance to subgroups
Subgroups
Filter...
None
Attrition
Department
EducationField
MaritalStatus
Order by relevance to variable
Display
Annotate
No comparison
Compare medians
Compare means
1470
- | 1470
Yes: 2.66 ± 0.8
No: 2.78 ± 0.7
2
3
1
1.50
2
2.50
3
WorkLifeBalance
Student's t: 2.178 (p=0.030, N=1470)
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share

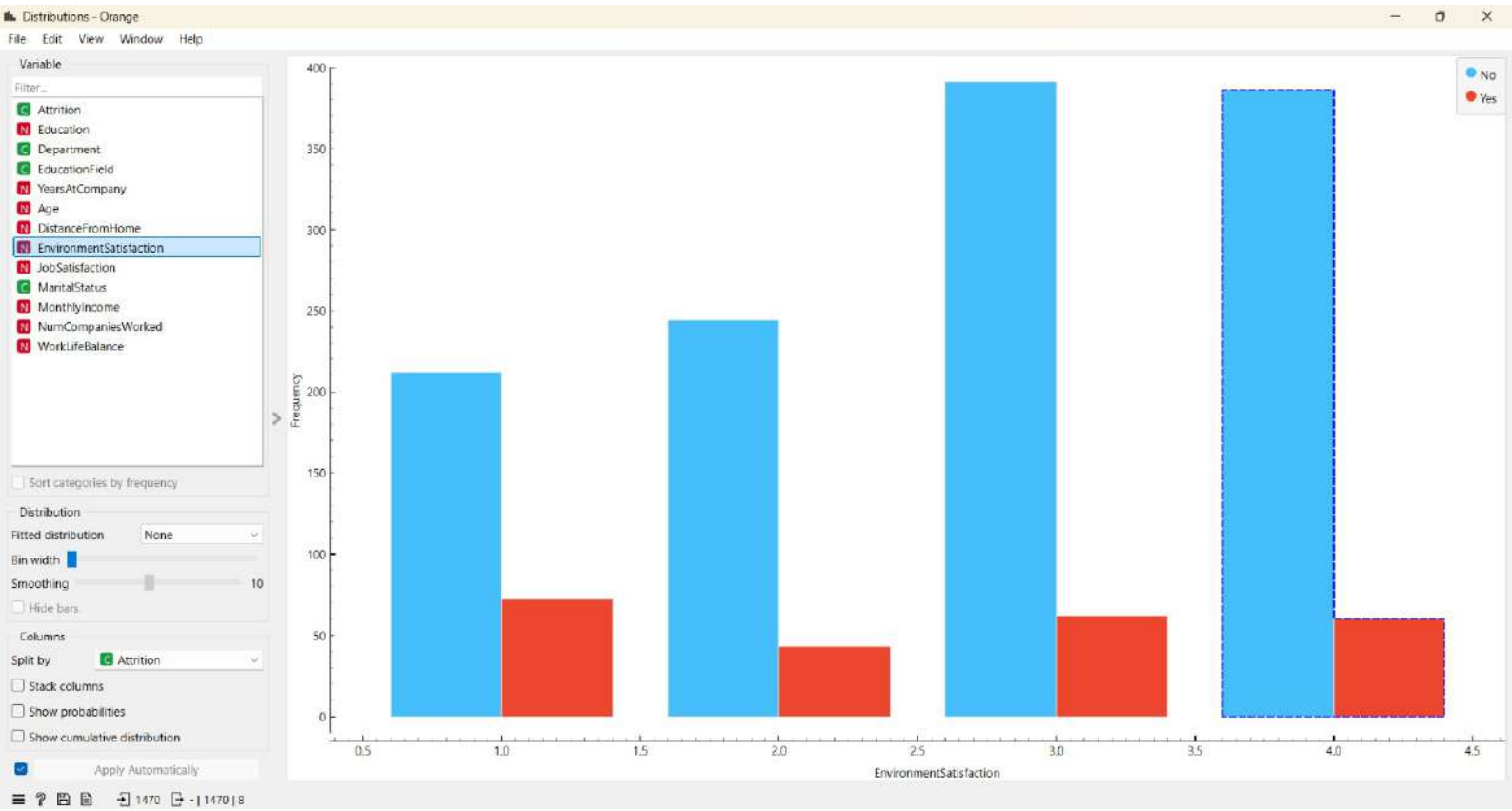

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 8
No
Yes
Frequency
0
100
200
300
400
500
600
700
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
WorkLifeBalance

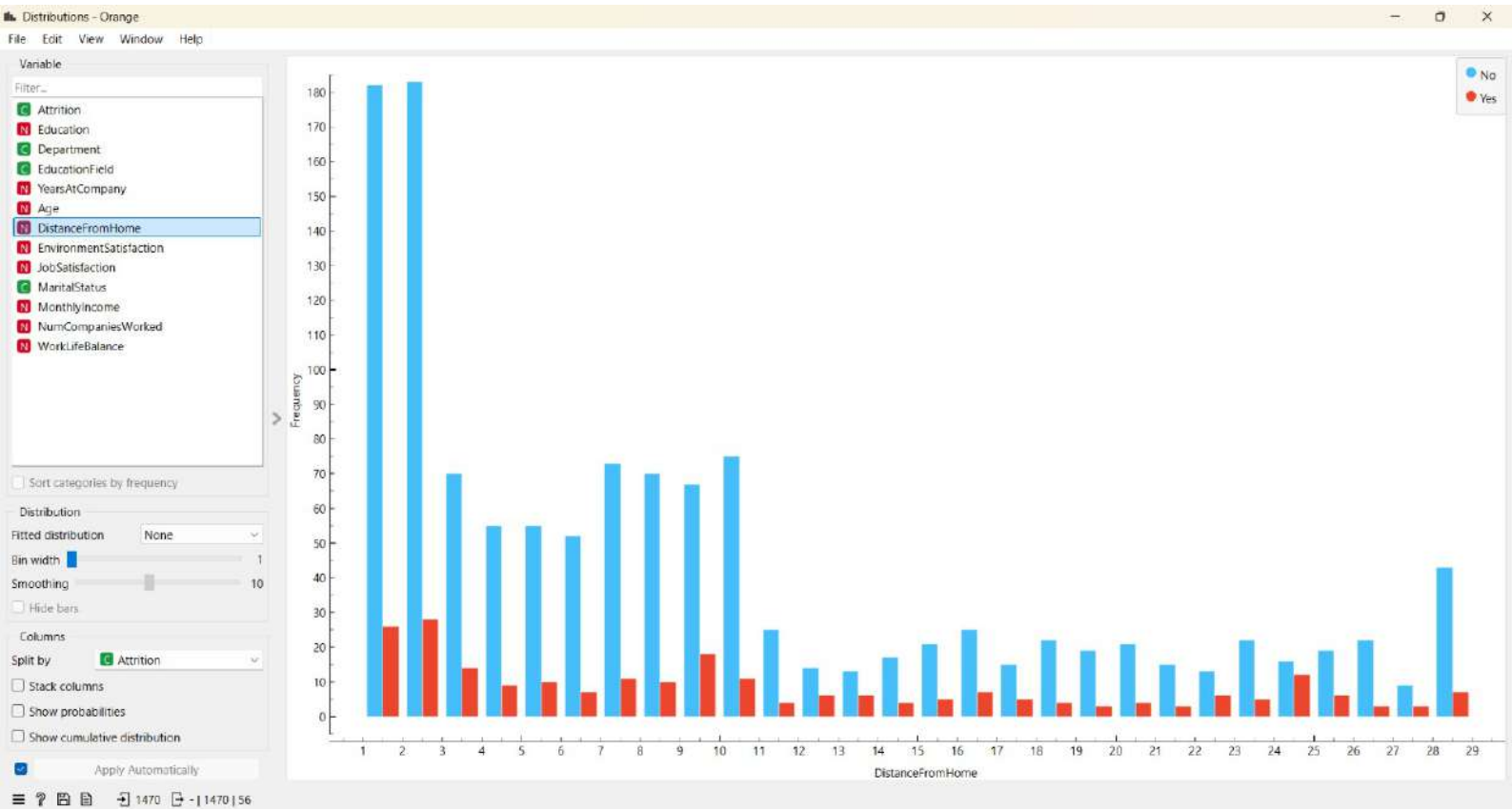

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 20
No
Yes
Frequency
0
50
100
150
200
250
300
350
400
0
1
2
3
4
5
6
7
8
9
NumCompaniesWorked

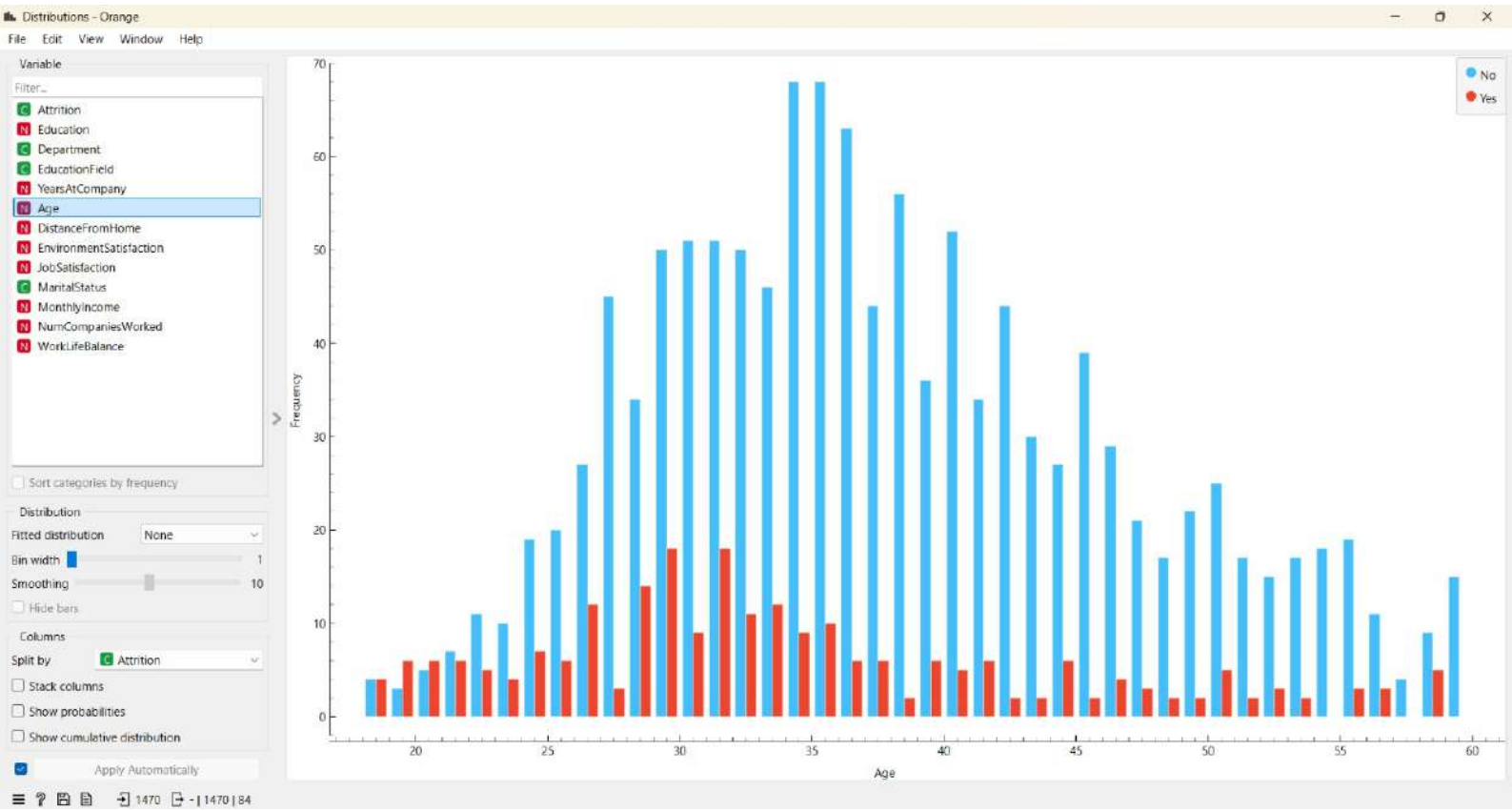

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width 500
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 76
Frequency
MonthlyIncome
No
Yes
0
20
40
60
80
100
120
2000
4000
6000
8000
10000
12000
14000
16000
18000
20000

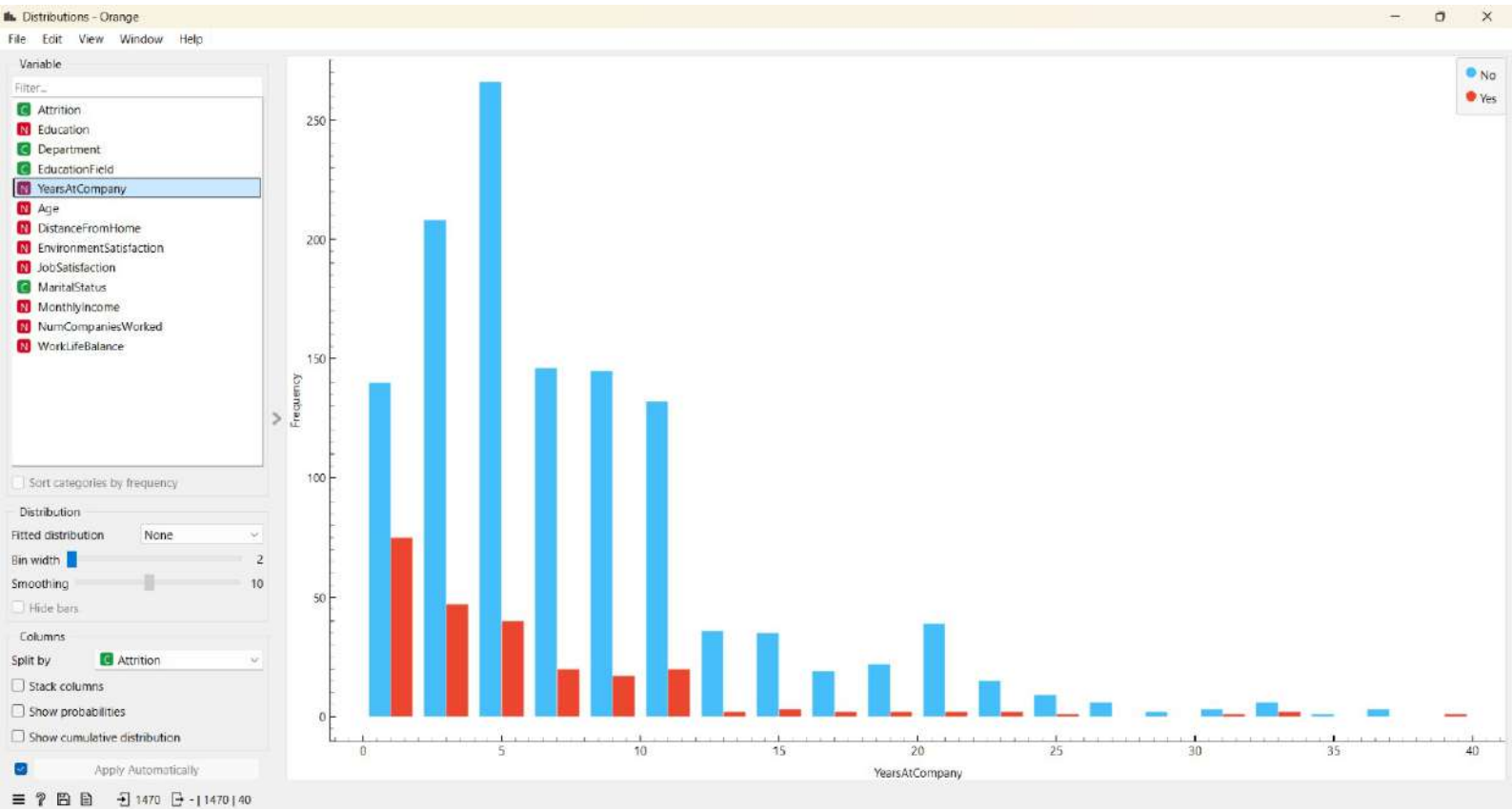

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 6
600
500
400
300
200
100
0
Frequency
No
Yes
Divorced
Married
Single
MaritalStatus

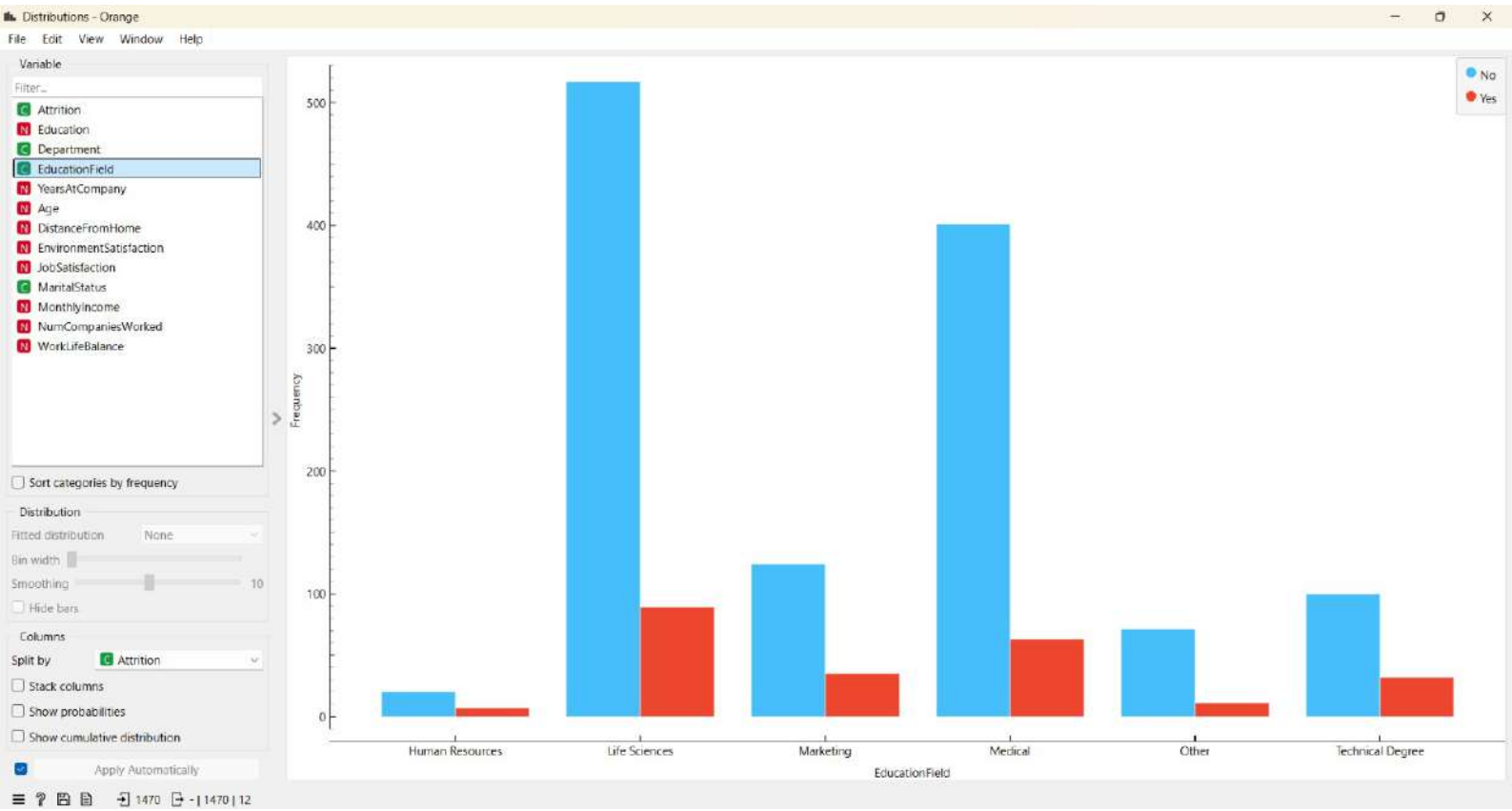

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 8
No
Yes
Frequency
0
50
100
150
200
250
300
350
400
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
JobSatisfaction

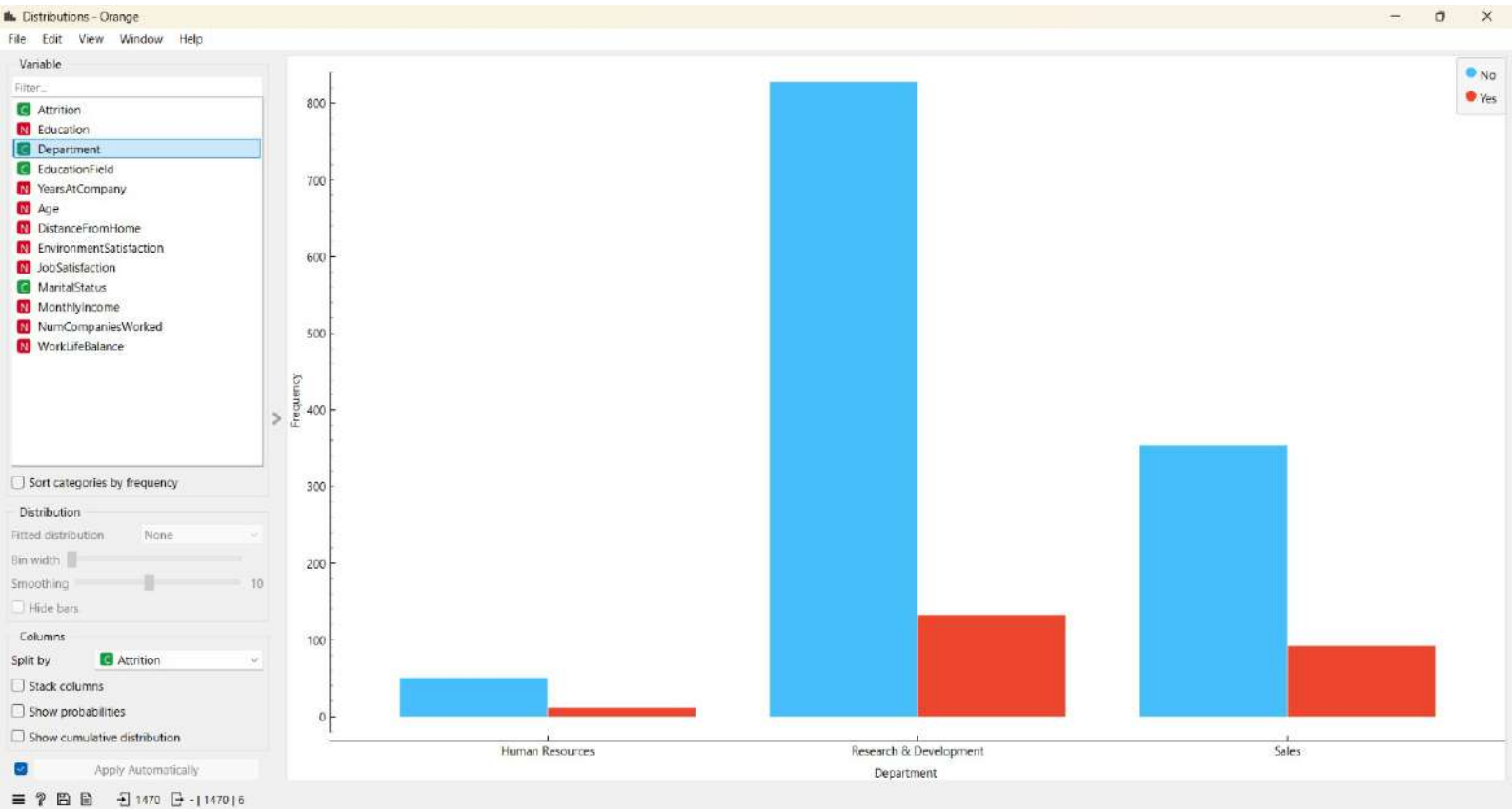

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 8
Frequency
EnvironmentSatisfaction
No
Yes

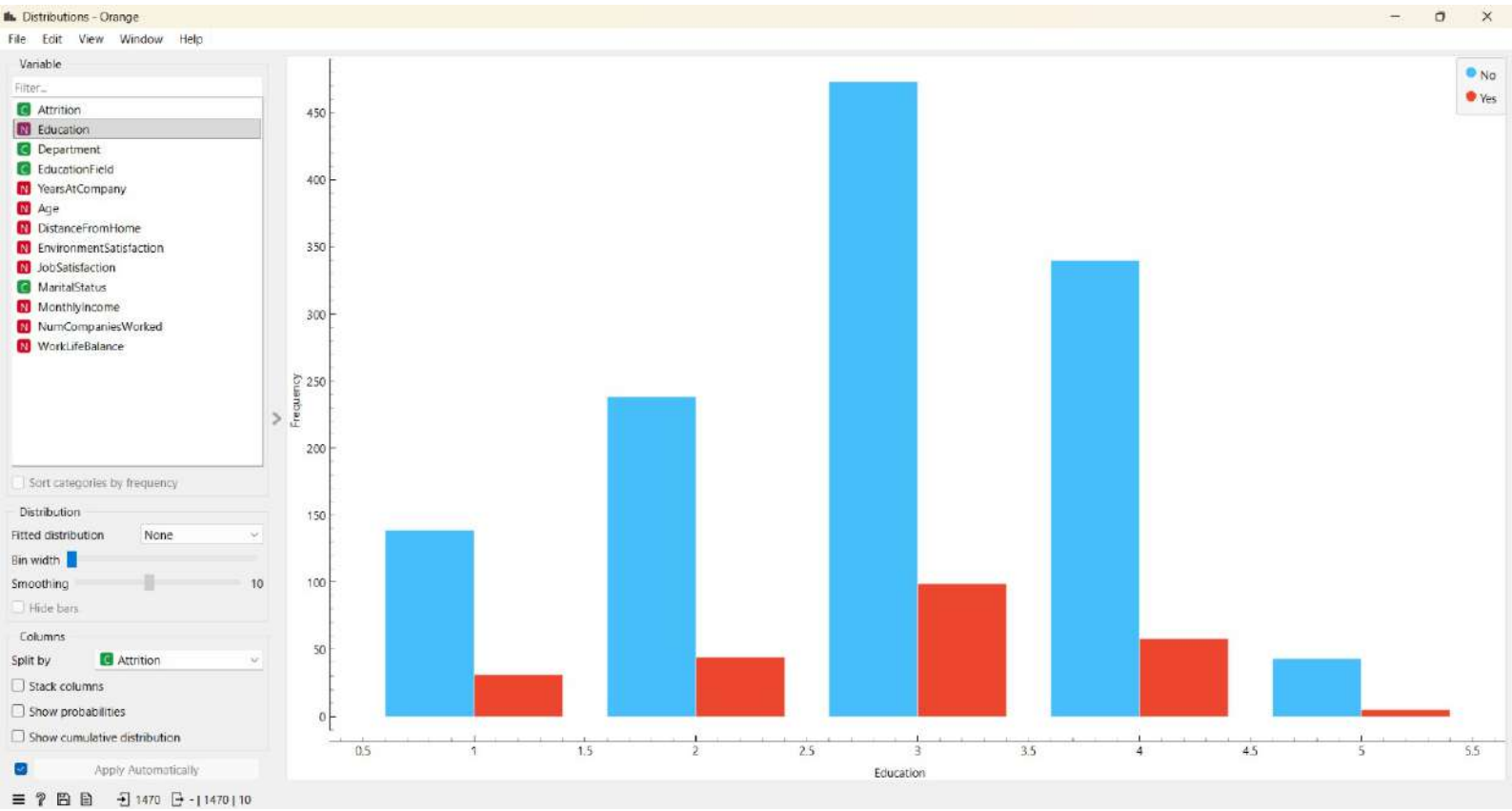

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width 1
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 56
No
Yes
Frequency
DistanceFromHome

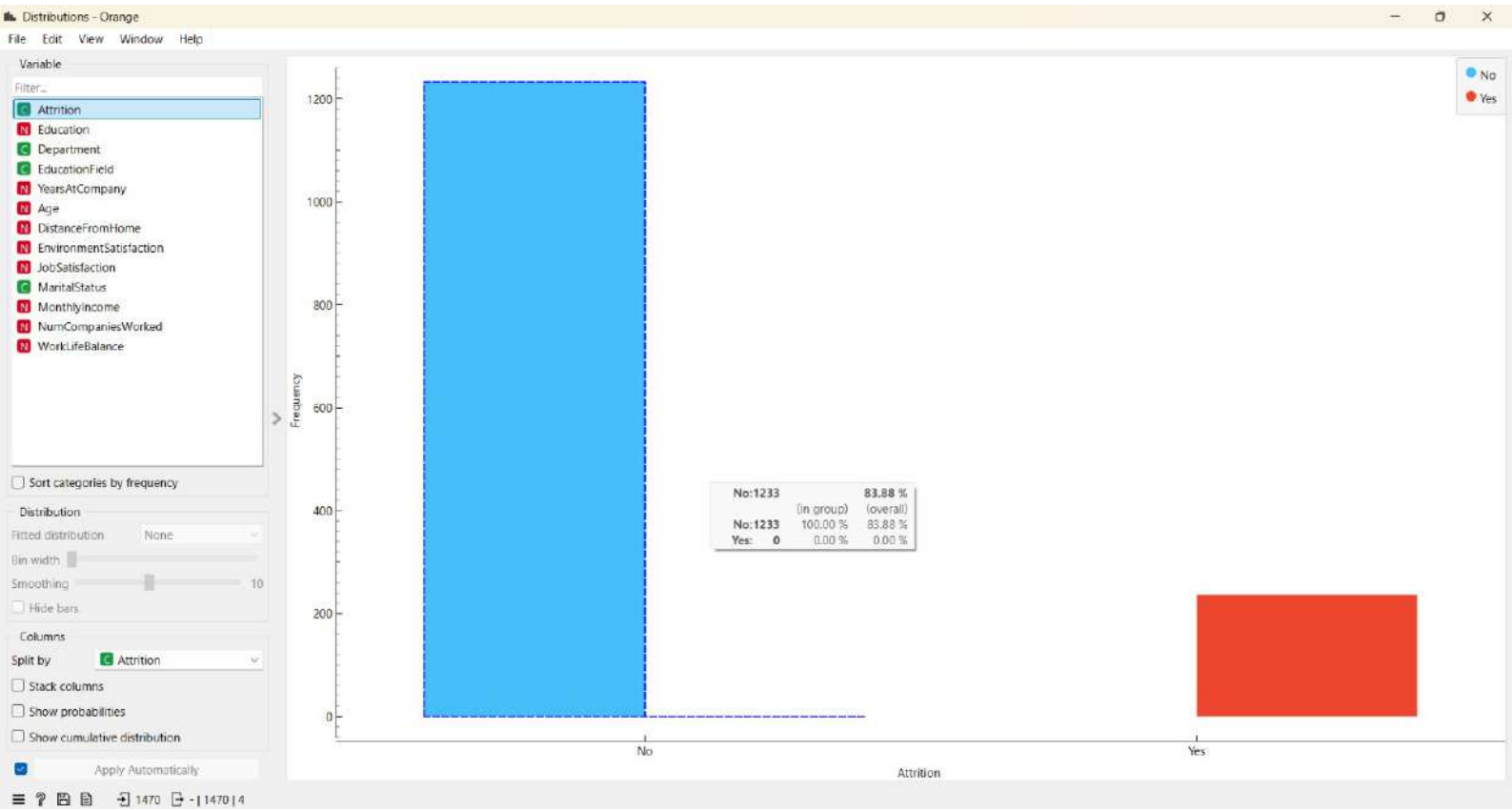

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width 1
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 84
Frequency
Age
No
Yes
0 10 20 30 40 50 60 70
20 25 30 35 40 45 50 55 60

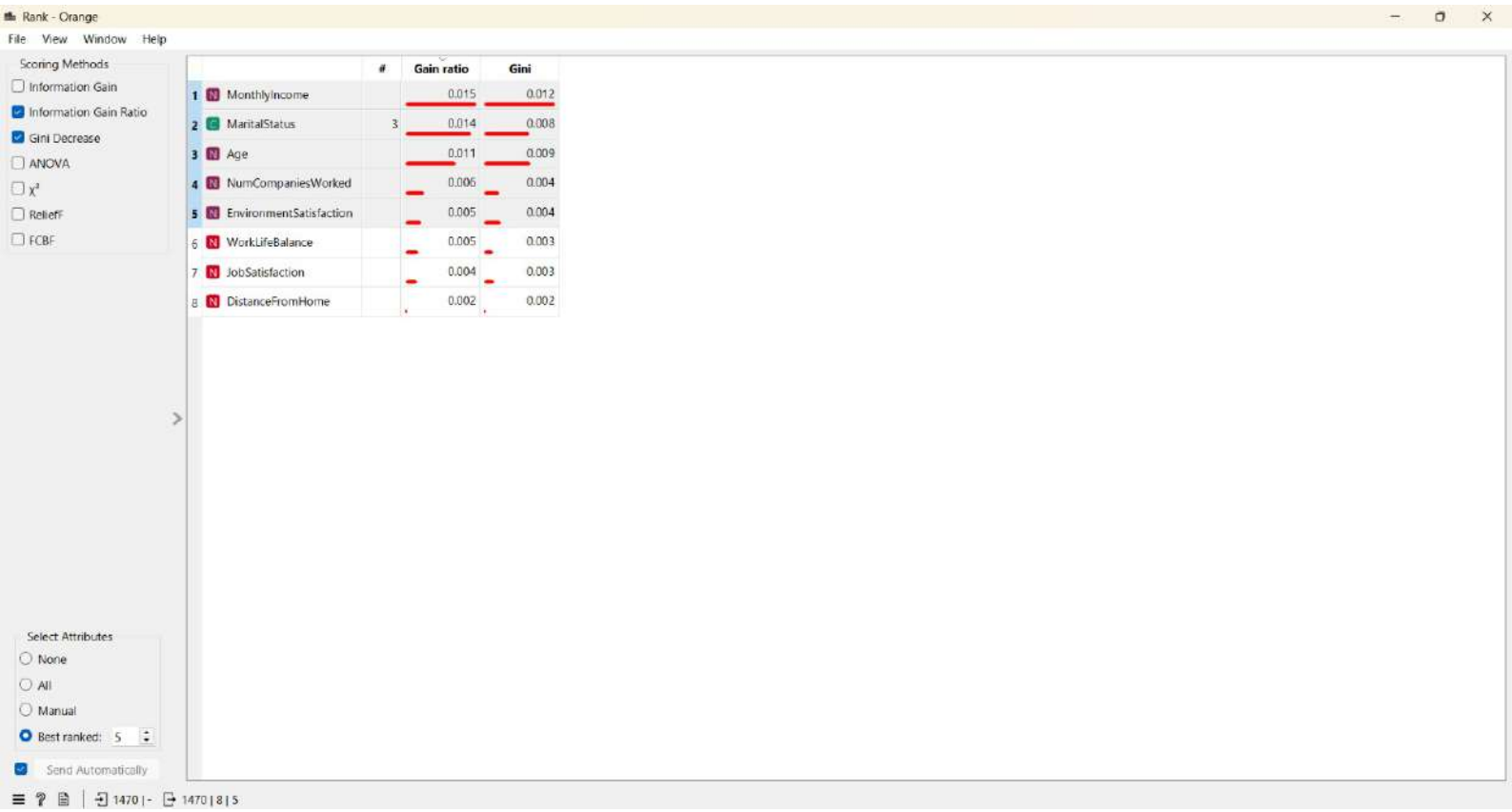

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width 2
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 40
No
Yes
Frequency
YearsAtCompany
0
50
100
150
200
250
0
5
10
15
20
25
30
35
40

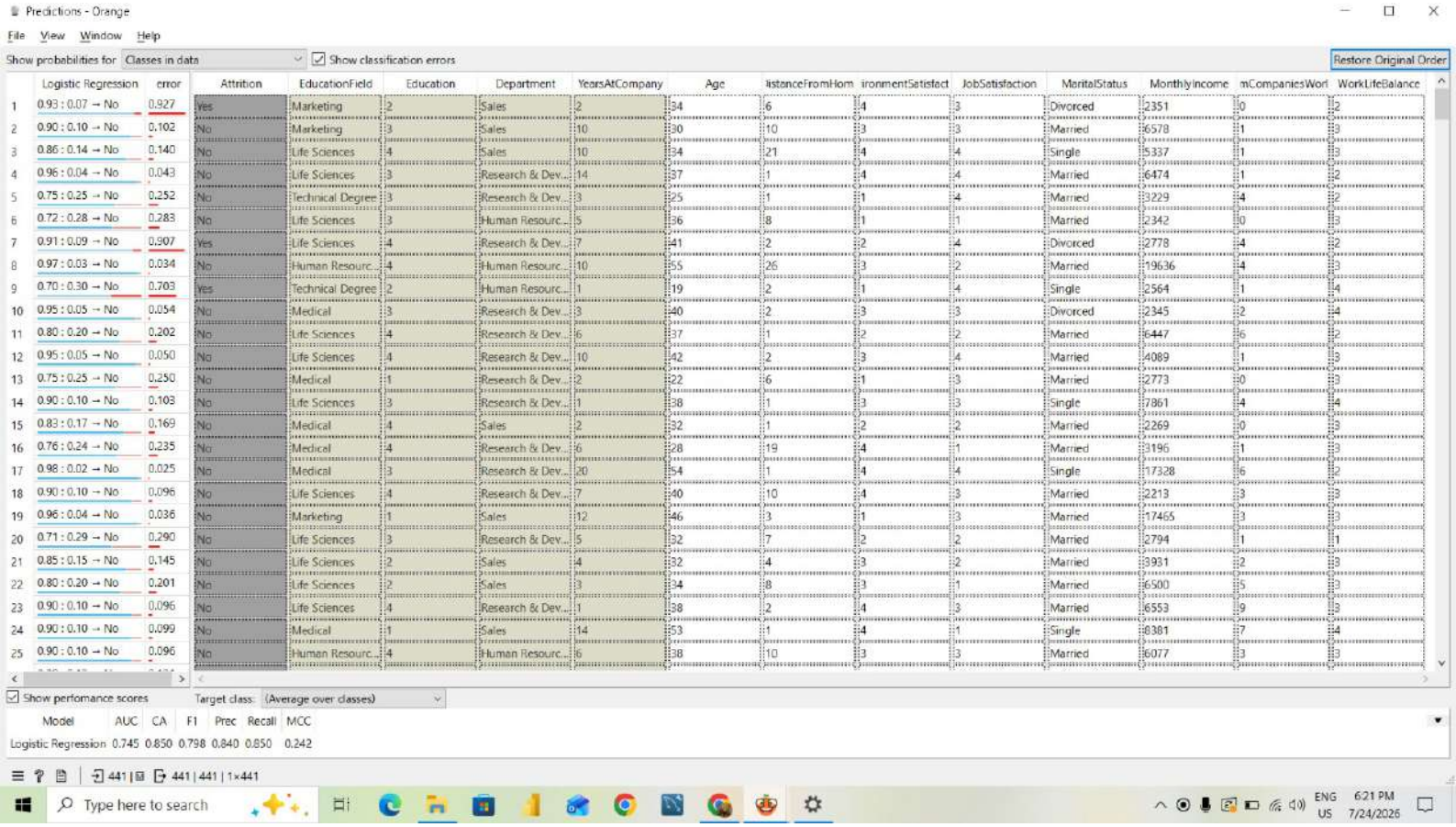

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 12
No
Yes
Frequency
0
100
200
300
400
500
Human Resources
Life Sciences
Marketing
Medical
Other
Technical Degree
EducationField

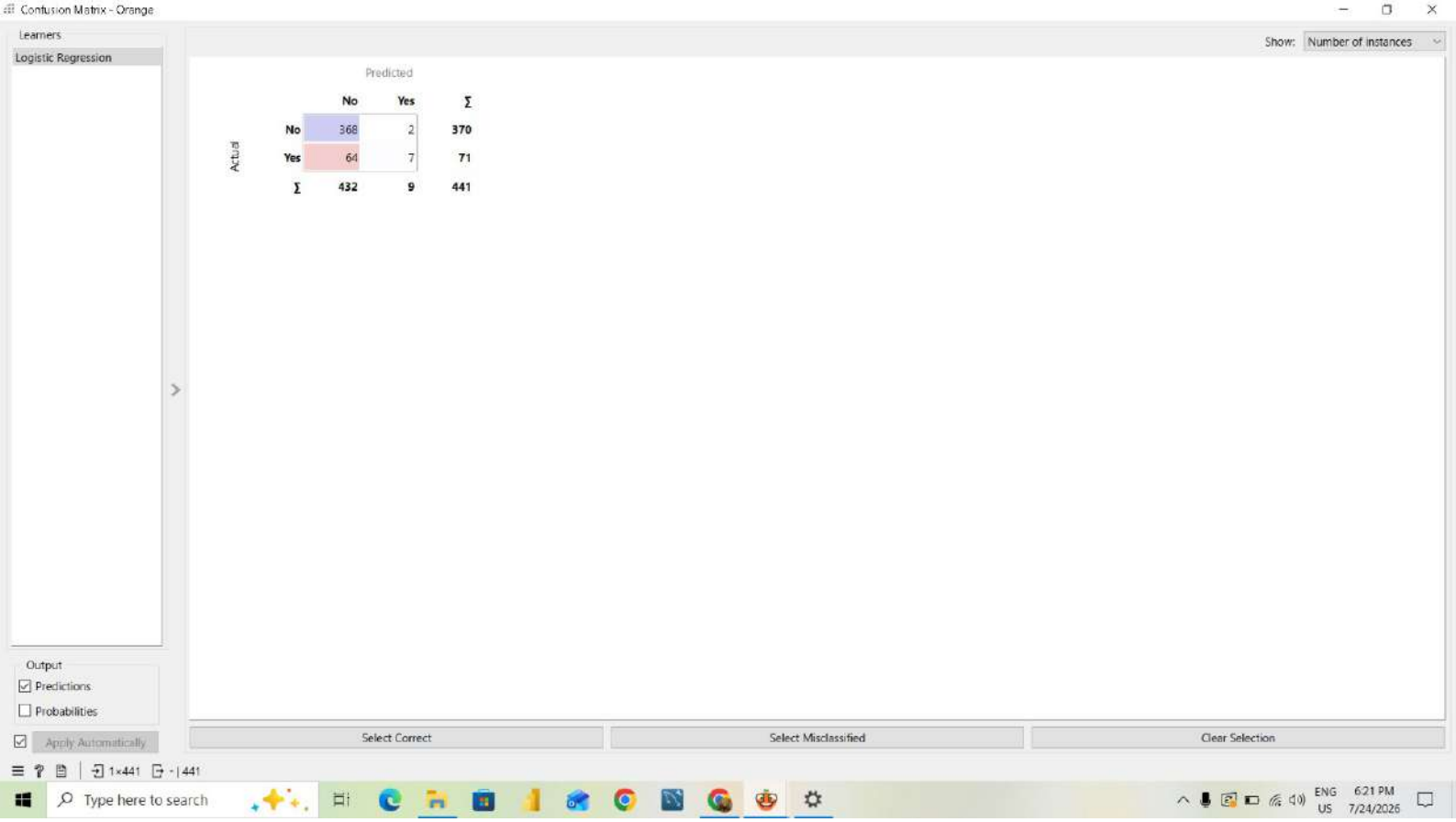

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 6
No
Yes
Frequency
800
700
600
500
400
300
200
100
0
Human Resources
Research & Development
Sales
Department

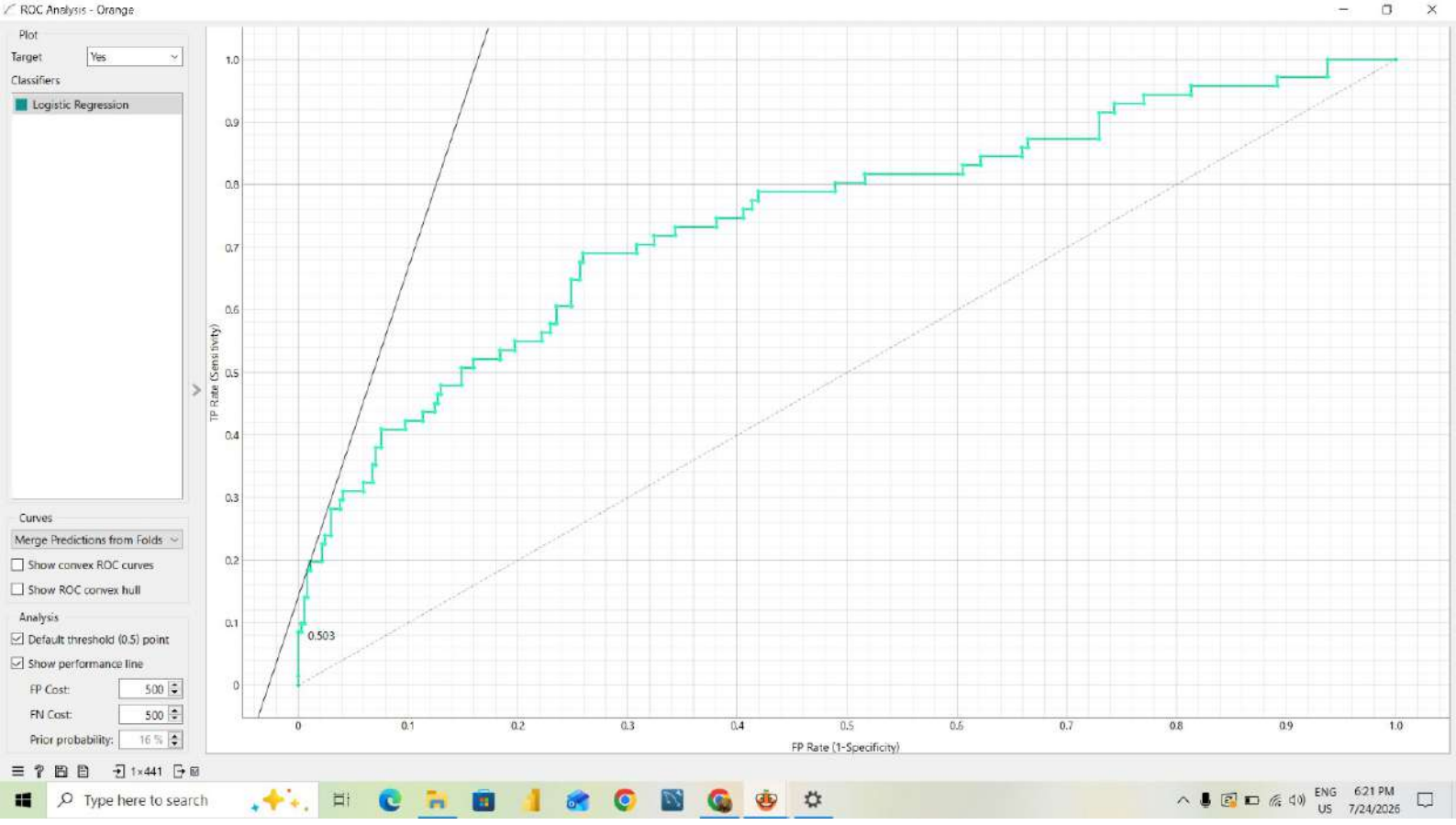

Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 10
No
Yes
Frequency
0
50
100
150
200
250
300
350
400
450
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
Education


Distributions - Orange
File Edit View Window Help
Variable
Filter...
Attrition
Education
Department
EducationField
YearsAtCompany
Age
DistanceFromHome
EnvironmentSatisfaction
JobSatisfaction
MaritalStatus
MonthlyIncome
NumCompaniesWorked
WorkLifeBalance
Sort categories by frequency
Distribution
Fitted distribution None
Bin width
Smoothing 10
Hide bars
Columns
Split by Attrition
Stack columns
Show probabilities
Show cumulative distribution
Apply Automatically
1470 - | 1470 | 4
No
Yes
Frequency
1200
1000
800
600
400
200
0
No:1233 83.88 %
(in group) (overall)
No:1233 100.00 % 83.88 %
Yes: 0 0.00 % 0.00 %
No
Yes
Attrition


Rank - Orange
File View Window Help
Scoring Methods
Information Gain
Information Gain Ratio
Gini Decrease
ANOVA
χ²
ReliefF
FCBF
Gain ratio Gini
1 MonthlyIncome 0.015 0.012
2 MaritalStatus 3 0.014 0.008
3 Age 0.011 0.009
4 NumCompaniesWorked 0.006 0.004
5 EnvironmentSatisfaction 0.005 0.004
6 WorkLifeBalance 0.005 0.003
7 JobSatisfaction 0.004 0.003
8 DistanceFromHome 0.002 0.002
Select Attributes
None
All
Manual
Best ranked: 5
Send Automatically
1470 | - 1470 | 8 | 5


Predictions - Orange
File View Window Help
Show probabilities for: Classes in data
Show classification errors
Restore Original Order
Logistic Regression | error | Attrition | EducationField | Education | Department | YearsAtCompany | Age | listanceFromHom | ironmentSatisfact | JobSatisfaction | MaritalStatus | MonthlyIncome | mCompaniesWorl | WorkLifeBalance
1 | 0.93 : 0.07 → No | 0.927 | Yes | Marketing | 2 | Sales | 2 | 34 | 6 | 4 | 3 | Divorced | 2351 | 0 | 2
2 | 0.90 : 0.10 → No | 0.102 | No | Marketing | 3 | Sales | 10 | 30 | 10 | 3 | 3 | Married | 6578 | 1 | 3
3 | 0.86 : 0.14 → No | 0.140 | No | Life Sciences | 4 | Sales | 10 | 34 | 21 | 4 | 4 | Single | 5337 | 1 | 3
4 | 0.96 : 0.04 → No | 0.043 | No | Life Sciences | 3 | Research & Dev... | 14 | 37 | 1 | 4 | 4 | Married | 6474 | 1 | 2
5 | 0.75 : 0.25 → No | 0.252 | No | Technical Degree | 3 | Research & Dev... | 3 | 25 | 1 | 1 | 4 | Married | 3229 | 4 | 2
6 | 0.72 : 0.28 → No | 0.283 | No | Life Sciences | 3 | Human Resourc... | 5 | 36 | 8 | 1 | 1 | Married | 2342 | 0 | 3
7 | 0.91 : 0.09 → No | 0.907 | Yes | Life Sciences | 4 | Research & Dev... | 7 | 41 | 2 | 2 | 4 | Divorced | 2778 | 4 | 2
8 | 0.97 : 0.03 → No | 0.034 | No | Human Resourc... | 4 | Human Resourc... | 10 | 55 | 26 | 3 | 2 | Married | 19636 | 4 | 3
9 | 0.70 : 0.30 → No | 0.703 | Yes | Technical Degree | 2 | Human Resourc... | 1 | 19 | 2 | 1 | 4 | Single | 2564 | 1 | 4
10 | 0.95 : 0.05 → No | 0.054 | No | Medical | 3 | Research & Dev... | 3 | 40 | 2 | 3 | 3 | Divorced | 2345 | 2 | 4
11 | 0.80 : 0.20 → No | 0.202 | No | Life Sciences | 4 | Research & Dev... | 6 | 37 | 1 | 2 | 2 | Married | 6447 | 6 | 2
12 | 0.95 : 0.05 → No | 0.050 | No | Life Sciences | 4 | Research & Dev... | 10 | 42 | 2 | 3 | 4 | Married | 4089 | 1 | 3
13 | 0.75 : 0.25 → No | 0.250 | No | Medical | 1 | Research & Dev... | 2 | 22 | 6 | 1 | 3 | Married | 2773 | 0 | 3
14 | 0.90 : 0.10 → No | 0.103 | No | Life Sciences | 3 | Research & Dev... | 1 | 38 | 1 | 3 | 3 | Single | 7861 | 4 | 4
15 | 0.83 : 0.17 → No | 0.169 | No | Medical | 4 | Sales | 2 | 32 | 1 | 2 | 2 | Married | 2269 | 0 | 3
16 | 0.76 : 0.24 → No | 0.235 | No | Medical | 4 | Research & Dev... | 6 | 28 | 19 | 4 | 1 | Married | 3196 | 1 | 3
17 | 0.98 : 0.02 → No | 0.025 | No | Medical | 3 | Research & Dev... | 20 | 54 | 1 | 4 | 4 | Single | 17328 | 6 | 2
18 | 0.90 : 0.10 → No | 0.096 | No | Life Sciences | 4 | Research & Dev... | 7 | 40 | 10 | 4 | 3 | Married | 2213 | 3 | 3
19 | 0.96 : 0.04 → No | 0.036 | No | Marketing | 1 | Sales | 12 | 46 | 3 | 1 | 3 | Married | 17465 | 3 | 3
20 | 0.71 : 0.29 → No | 0.290 | No | Life Sciences | 3 | Research & Dev... | 5 | 32 | 7 | 2 | 2 | Married | 2794 | 1 | 1
21 | 0.85 : 0.15 → No | 0.145 | No | Life Sciences | 2 | Sales | 4 | 32 | 4 | 3 | 2 | Married | 3931 | 2 | 3
22 | 0.80 : 0.20 → No | 0.201 | No | Life Sciences | 2 | Sales | 3 | 34 | 8 | 3 | 1 | Married | 6500 | 5 | 3
23 | 0.90 : 0.10 → No | 0.096 | No | Life Sciences | 4 | Research & Dev... | 1 | 38 | 2 | 4 | 3 | Married | 6553 | 9 | 3
24 | 0.90 : 0.10 → No | 0.099 | No | Medical | 1 | Sales | 14 | 53 | 1 | 4 | 1 | Single | 8381 | 7 | 4
25 | 0.90 : 0.10 → No | 0.096 | No | Human Resourc... | 4 | Human Resourc... | 6 | 38 | 10 | 3 | 3 | Married | 6077 | 3 | 3
Show performance scores
Target class: (Average over classes)
Model | AUC | CA | F1 | Prec | Recall | MCC
Logistic Regression 0.745 0.850 0.798 0.840 0.850 0.242
441 | 441 | 441 | 1×441
Type here to search
ENG US
6:21 PM
7/24/2026


Confusion Matrix - Orange
Learners
Logistic Regression
Show:
Number of instances
Predicted
No
Yes
Σ
Actual
No
368
2
370
Yes
64
7
71
Σ
432
9
441
Output
Predictions
Probabilities
Apply Automatically
Select Correct
Select Misclassified
Clear Selection
1×441
-|441
Type here to search
ENG
US
6:21 PM
7/24/2026


ROC Analysis - Orange
Plot
Target
Yes
Classifiers
Logistic Regression
Curves
Merge Predictions from Folds
Show convex ROC curves
Show ROC convex hull
Analysis
Default threshold (0.5) point
Show performance line
FP Cost:
500
FN Cost:
500
Prior probability:
16 %
1×441
0.503
TP Rate (Sensitivity)
FP Rate (1-Specificity)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

Output — PSPPIRE Output Viewer

File Edit Windows Help

- Get Data
- Variable Alignr
- Variable Widt
- Variable Alignr
- Variable Widt
- Variable Alignr
- Variable Widt
- Variable Alignr
- Logistic Regre

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	1110.95	.12	.20

Categorical Variables' Codings

		Frequency	Parameter coding (1)	(2)	(3)	(4)	(5)
Department	Human Resources	63	1	0			
	Research & Development	961	0	1			
	Sales	446	0	0			
EducationField	Human Resources	27	1	0	0	0	0
	Life Sciences	606	0	1	0	0	0
	Marketing	159	0	0	1	0	0
	Medical	464	0	0	0	1	0
	Other	82	0	0	0	0	1
	Technical Degree	132	0	0	0	0	0
MaritalStatus	Divorced	327	1	0			
	Married	673	0	1			
	Single	470	0	0			

Classification Table

	Observed		Predicted: Attrition Yes	Attrition No	Percentage Correct
Step 1	Attrition	Yes	96	201	15.2%
		No	7	1226	99.4%
	Overall Percentage				85.9%

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1	Age	.04	.01	10.72	1	.001	1.04
	DistanceFromHo	-.03	.01	11.92	1	.001	.97
	Education	.03	.08	.13	1	.720	1.03
	EnvironmentSat	.30	.07	17.93	1	.000	1.35
	JobSatisfaction	.29	.07	17.36	1	.000	1.33
	MonthlyIncome	.00	2.87E-005	13.15	1	.000	1.00
	NumCompanies	-.12	.03	13.21	1	.000	.89
	WorkLifeBalanc	.28	.11	7.10	1	.008	1.33
	YearsAtCompan	.02	.02	1.29	1	.255	1.02
	Department			6.46	2	.039	
	Department(1)	.35	.54	.42	1	.515	1.42
	Department(2)	.49	.19	6.46	1	.011	1.64
	EducationField			11.47	5	.043	
	EducationField(	-.31	.73	.18	1	.668	.73
	EducationField(	.61	.26	5.75	1	.016	1.85
	EducationField(	.26	.34	.61	1	.435	1.30
	EducationField(	.78	.27	8.46	1	.004	2.18
	EducationField(	.65	.41	2.56	1	.109	1.92
	MaritalStatus			38.48	2	.000	
	MaritalStatus(1)	1.15	.23	25.78	1	.000	3.16
	MaritalStatus(2)	.88	.17	26.72	1	.000	2.40
	Constant	-3.50	.61	32.47	1	.000	.03

Type here to search

ENG IN 11:14 AM 7/24/2026

File Edit Windows Help

- Get Data
- Variable Align
- Variable Widt
- Variable Align
- Variable Widt
- Variable Align
- Variable Widt
- Variable Align
- Correlations

```
    2 |         /VARIABLES =  Age Attrition Department DistanceFromHome Education EducationField EnvironmentSatisfaction
JobSatisfaction MaritalStatus MonthlyIncome NumCompaniesWorked WorkLifeBalance YearsAtCompany

!
~~~~~~~~~~~~

        /VARIABLES =  Age Attrition Department
DistanceFromHome Education EducationField
EnvironmentSatisfaction JobSatisfaction
MaritalStatus MonthlyIncome NumCompaniesWorked
WorkLifeBalance YearsAtCompany
        /PRINT = TWOTAIL SIG.
```

Correlations

		Age	nceFromH	Education	nmentSatisf	JobSatisfaction	nthlyInco	mpaniesW	rkLifeBalan	rsAtCompa
Age	Pearson Correlation	1.000	-.002	.208	.010	-.005	.498	.300	-.021	.311
	Sig. (2-tailed)		.948	.000	.697	.851	.000	.000	.410	.000
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
DistanceFromHo	Pearson Correlation	-.002	1.000	.021	-.016	-.004	-.017	-.029	-.027	.010
	Sig. (2-tailed)	.948		.420	.538	.888	.515	.262	.309	.716
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
Education	Pearson Correlation	.208	.021	1.000	-.027	-.011	.095	.126	.010	.069
	Sig. (2-tailed)	.000	.420		.299	.665	.000	.000	.707	.008
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
EnvironmentSat	Pearson Correlation	.010	-.016	-.027	1.000	-.007	-.006	.013	.028	.001
	Sig. (2-tailed)	.697	.538	.299		.795	.811	.629	.290	.955
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
JobSatisfaction	Pearson Correlation	-.005	-.004	-.011	-.007	1.000	-.007	-.056	-.019	-.004
	Sig. (2-tailed)	.851	.888	.665	.795		.784	.033	.456	.884
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
MonthlyIncome	Pearson Correlation	.498	-.017	.095	-.006	-.007	1.000	.150	.031	.514
	Sig. (2-tailed)	.000	.515	.000	.811	.784		.000	.240	.000
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
NumCompanies	Pearson Correlation	.300	-.029	.126	.013	-.056	.150	1.000	-.008	-.118
	Sig. (2-tailed)	.000	.262	.000	.629	.033	.000		.749	.000
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
WorkLifeBalanc	Pearson Correlation	-.021	-.027	.010	.028	-.019	.031	-.008	1.000	.012
	Sig. (2-tailed)	.410	.309	.707	.290	.456	.240	.749		.643
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470
YearsAtCompan	Pearson Correlation	.311	.010	.069	.001	-.004	.514	-.118	.012	1.000
	Sig. (2-tailed)	.000	.716	.008	.955	.884	.000	.000	.643	
	N	1470	1470	1470	1470	1470	1470	1470	1470	1470